2.5.38 Graph Connectivity: GATE IT 2007 | Question: 25

What is the largest integer m such that every simple connected graph with n vertices and n edges contains at least m different spanning trees ?

A. 1 B. 2 C. 3 D. n

gateit-2007 graph-theory graph-connectivity tricky

Answer key

2.5.39 Graph Connectivity: GATE IT 2008 | Question: 27

G is a simple undirected graph. Some vertices of G are of odd degree. Add a node v to G and make it adjacent to each odd degree vertex of G. The resultant graph is sure to be

A. regular B. complete C. Hamiltonian D. Euler

gateit-2008 graph-theory graph-connectivity normal

Answer key

2.5.40 Graph Connectivity: GATE IT 2008 | Question: 4

What is the size of the smallest MIS (Maximal Independent Set) of a chain of nine nodes?

A. 5 B. 4 C. 3 D. 2

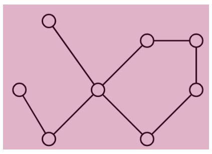
gateit-2008 normal graph-connectivity

Answer key

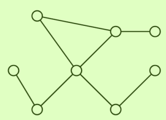
2.6 Graph Isomorphism (4)

2.6.1 Graph Isomorphism: GATE CSE 2012 | Question: 26

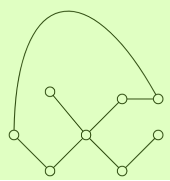
Which of the following graphs is isomorphic to



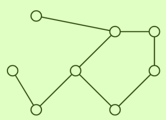
A.



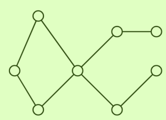
B.



C.



D.



gatecse-2012 graph-theory graph-isomorphism normal

Answer key

2.6.2 Graph Isomorphism: GATE CSE 2014 | Set 2 | Question: 51

A cycle on n vertices is isomorphic to its complement. The value of n is _____.

gatecse-2014-set2 graph-theory numerical-answers normal graph-isomorphism

Answer key

2.6.3 Graph Isomorphism: GATE CSE 2015 | Set 2 | Question: 28

A graph is self-complementary if it is isomorphic to its complement. For all self-complementary graphs on n vertices, n is

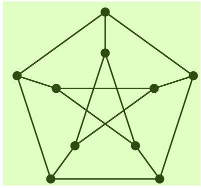
A. A multiple of 4
B. Even
C. Odd
D. Congruent to 0 mod 4, or, 1 mod 4.

gatecse-2015-set2 graph-theory graph-isomorphism

Answer key

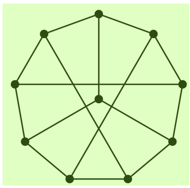
2.6.4 Graph Isomorphism: GATE CSE 2022 | Question: 40

The following simple undirected graph is referred to as the Peterson graph.



Which of the following statements is/are TRUE?

A. The chromatic number of the graph is 3.
B. The graph has a Hamiltonian path.
C. The following graph is isomorphic to the Peterson graph.



D. The size of the largest independent set of the given graph is 3. (A subset of vertices of a graph form an independent set if no two vertices of the subset are adjacent.)

gatecse-2022 graph-theory graph-isomorphism multiple-selects two-marks

Answer key

2.7 Graph Matching (2)

2.7.1 Graph Matching: GATE CSE 2003 | Question: 36

How many perfect matching are there in a complete graph of 6 vertices?

A. 15
B. 24
C. 30
D. 60

gatecse-2003 graph-theory graph-matching normal

Answer key

2.7.2 Graph Matching: GATE CSE 2026 | Set 1 | Question: 47

Let G be an undirected graph, which is a path on 8 vertices. The number of matchings in G is _____. (answer in integer)

gatecse-2026-set1 graph-theory graph-matching numerical-answers two-marks

Answer key

2.8 Graph Planarity (13)

2.8.1 Graph Planarity: GATE CSE 1987 | Question: 2e

State whether the following statement is TRUE or FALSE:

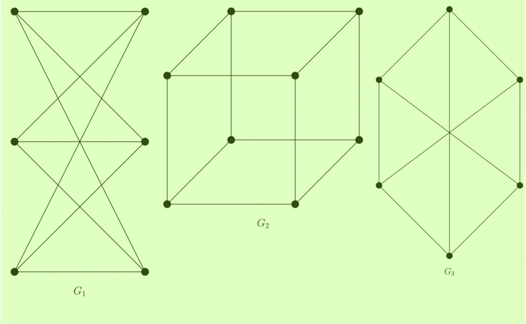
There is a linear-time algorithm for testing the planarity of finite graphs.

gate1987 graph-theory graph-planarity true-false

Answer key

2.8.2 Graph Planarity: GATE CSE 1989 | Question: 3-vi

Which of the following graphs is/are planar?



gatel989 normal graph-theory graph-planarity descriptive

Answer key

2.8.3 Graph Planarity: GATE CSE 1990 | Question: 3-xi

A graph is planar if and only if,

A. It does not contain a subgraph homeomorphic to k_5 and $k_{3,3}$.
B. It does not contain a subgraph isomorphic to k_5 and $k_{3,3}$.
C. It does not contain a subgraph isomorphic to k_5 or $k_{3,3}$
D. It does not contain a subgraph homeomorphic to k_5 or $k_{3,3}$.

gate1990 normal graph-theory graph-planarity multiple-selects

Answer key

2.8.4 Graph Planarity: GATE CSE 1992 | Question: 01,x

Maximum number of edges in a planar graph with n vertices is _____

gate1992 graph-theory graph-planarity easy fill-in-the-blanks

Answer key

2.8.5 Graph Planarity: GATE CSE 1992 | Question: 02,viii

A non-planar graph with minimum number of vertices has

A. 9 edges, 6 vertices
B. 6 edges, 4 vertices
C. 10 edges, 5 vertices
D. 9 edges, 5 vertices

gate1992 graph-theory normal graph-planarity

Answer key

2.8.6 Graph Planarity: GATE CSE 2005 | Question: 10

Let G be a simple connected planar graph with 13 vertices and 19 edges. Then, the number of faces in the planar embedding of the graph is:

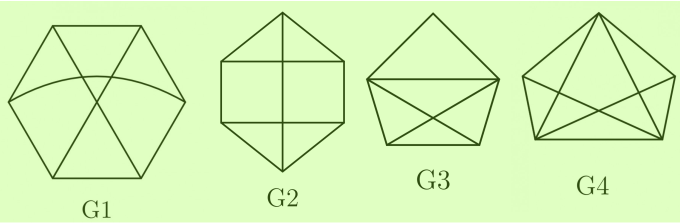
A. 6
B. 8
C. 9
D. 13

gatecse-2005 graph-theory graph-planarity

Answer key

2.8.7 Graph Planarity: GATE CSE 2005 | Question: 47

Which one of the following graphs is NOT planar?



A. G1 B. G2 C. G3 D. G4

gatecse-2005 graph-theory graph-planarity normal

Answer key

2.8.8 Graph Planarity: GATE CSE 2008 | Question: 23

Which of the following statements is true for every planar graph on n vertices?

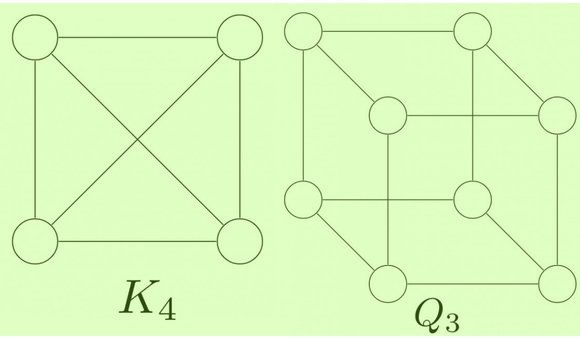
A. The graph is connected
B. The graph is Eulerian
C. The graph has a vertex-cover of size at most $\frac{3n}{4}$
D. The graph has an independent set of size at least $\frac{n}{3}$

gatecse-2008 graph-theory normal graph-planarity

Answer key

2.8.9 Graph Planarity: GATE CSE 2011 | Question: 17

K4 and Q3 are graphs with the following structures.



Which one of the following statements is **TRUE** in relation to these graphs?

A. K4 is a planar while Q3 is not
B. Both K4 and Q3 are planar
C. Q3 is planar while K4 is not
D. Neither K4 nor Q3 is planar

gatecse-2011 graph-theory graph-planarity normal

Answer key

2.8.10 Graph Planarity: GATE CSE 2012 | Question: 17

Let G be a simple undirected planar graph on 10 vertices with 15 edges. If G is a connected graph, then the number of **bounded** faces in any embedding of G on the plane is equal to

A. 3 B. 4 C. 5 D. 6

gatecse-2012 graph-theory graph-planarity normal

Answer key

2.8.11 Graph Planarity: GATE CSE 2014 | Set 3 | Question: 52

Let δ denote the minimum degree of a vertex in a graph. For all planar graphs on n vertices with $\delta \geq 3$, which one of the following is **TRUE**?

A. In any planar embedding, the number of faces is at least $\frac{n}{2} + 2$

B. In any planar embedding, the number of faces is less than $\frac{n}{2}+2$
C. There is a planar embedding in which the number of faces is less than $\frac{n}{2}+2$
D. There is a planar embedding in which the number of faces is at most $\frac{n}{\delta+1}$

gatecse-2014-set3 graph-theory graph-planarity normal

Answer key

2.8.12 Graph Planarity: GATE CSE 2015 | Set 1 | Question: 54

Let G be a connected planar graph with 10 vertices. If the number of edges on each face is three, then the number of edges in G is_______________.

gatecse-2015-set1 graph-theory graph-connectivity normal graph-planarity numerical-answers

Answer key

2.8.13 Graph Planarity: GATE CSE 2021 | Set 1 | Question: 16

In an undirected connected planar graph G, there are eight vertices and five faces. The number of edges in G is __________.

gatecse-2021-set1 graph-theory graph-planarity numerical-answers easy one-mark

Answer key

2.9 Jaccard Coefficient (1)

2.9.1 Jaccard Coefficient: GATE CSE 2026 | Set 2 | Question: 26

Consider a complete graph K_n with n vertices ($n>4$). Note that multiple spanning trees can be constructed over K_n. Each of these spanning trees is represented as a set of edges. The Jaccard coefficient between any two sets is defined as the ratio of the size of the intersection of the two sets to the size of the union of the two sets.

Which one of the following options gives the lowest possible value for the Jaccard coefficient between any two spanning trees of K_n ?

A. $\frac{1}{n}$ B. $\frac{1}{2n-3}$ C. 0 D. $\frac{1}{n-1}$

gatecse-2026-set2 graph-theory jaccard-coefficient two-marks

Answer key

Answer Keys

2.1.1	D	2.1.2	D	2.1.3	X	2.2.1	N/A	2.2.2	N/A
2.2.3	N/A	2.2.4	D	2.2.5	C	2.2.6	C	2.2.7	B
2.2.8	C	2.2.9	D	2.2.10	C	2.2.11	C	2.2.12	16
2.2.13	A	2.3.1	B;C	2.4.1	D	2.4.2	C	2.4.3	A
2.4.4	4	2.4.5	3	2.4.6	7	2.4.7	A;B	2.4.8	B;C
2.4.9	2	2.4.10	C	2.4.11	B	2.5.1	N/A	2.5.2	N/A
2.5.3	N/A	2.5.4	N/A	2.5.5	N/A	2.5.6	C;D	2.5.7	C
2.5.8	N/A	2.5.9	N/A	2.5.10	B	2.5.11	D	2.5.12	N/A
2.5.13	B	2.5.14	C	2.5.15	B	2.5.16	A	2.5.17	B
2.5.18	C	2.5.19	D	2.5.20	A	2.5.21	506	2.5.22	36
2.5.23	C	2.5.24	B	2.5.25	C;D	2.5.26	C	2.5.27	A
2.5.28	36	2.5.29	D	2.5.30	A	2.5.31	16	2.5.32	D
2.5.33	B	2.5.34	D	2.5.35	A	2.5.36	B	2.5.37	D

2.5.38	C	2.5.39	D	2.5.40	C	2.6.1	B	2.6.2	5	1
2.6.3	D	2.6.4	A;B;C	2.7.1	A	2.7.2	34 : 34	2.8.1	TBA	
2.8.2	N/A	2.8.3	D	2.8.4	N/A	2.8.5	C	2.8.6	B	
2.8.7	A	2.8.8	C	2.8.9	B	2.8.10	D	2.8.11	A	
2.8.12	24	2.8.13	11 : 11	2.9.1	C					

3 Discrete Mathematics: Mathematical Logic (78)

Syllabus: Propositional and first order logic.

Mark Distribution in Previous GATE

Year	2026 - 1	2026 - 2	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum
1 Mark Count	0	1	0	1	0	1	1	0	1	1	0
2 Marks Count	0	0	1	0	0	0	0	0	0	0	0
Total Marks	0	1	2	1	0	1	1	0	1	1	**0**

Welcome to the "Discrete Mathematics: Mathematical Logic" chapter of your GATE Computer Science exam preparation. This section is designed to provide a comprehensive, exam-focused overview of the fundamental concepts, formulas, and problem-solving techniques you'll need to master. Mathematical Logic is the bedrock of computer science, providing the formal tools to reason about computation, verify program correctness, design circuits, and understand artificial intelligence. For the GATE CS exam, this subject typically carries a weightage of 5-8 marks. Questions often involve determining the validity of arguments, checking logical equivalence, translating natural language statements into logical expressions, and identifying tautologies or contradictions. A strong grasp of these concepts is indispensable not just for direct questions but also for understanding the logical underpinnings of other topics like set theory, relations, and algorithms.

Topic-wise Key Concepts

Propositional Logic

Propositional Logic, also known as propositional calculus, is the study of propositions (declarative sentences that are either true or false, but not both) and their combinations using logical connectives. It forms the most basic level of logical reasoning, focusing on the truth values of simple statements and how they combine to determine the truth values of complex statements.

- **Definition and Core Idea:** A proposition is a statement that is unequivocally true or false. Propositional logic uses symbols to represent propositions and logical connectives to form compound propositions, allowing us to analyze their truth values systematically.
- **Important Formulas, Theorems, and Results:**
  1. **Logical Connectives:**
     - Negation (NOT): $\neg p$ (read as "not p")
     - Conjunction (AND): $p \land q$ (read as "p and q")
     - Disjunction (OR): $p \lor q$ (read as "p or q")
     - Implication (IF...THEN): $p \to q$ (read as "if p then q" or "p implies q")
     - Biconditional (IF AND ONLY IF): $p \leftrightarrow q$ (read as "p if and only if q")
     - Exclusive OR (XOR): $p \oplus q$ (read as "p xor q")
     - NAND: $p \uparrow q \equiv \neg(p \land q)$
     - NOR: $p \downarrow q \equiv \neg(p \lor q)$
  2. **Truth Tables:** Essential for defining connectives and evaluating compound propositions.
     - $\neg p$: T if p is F, F if p is T.
     - $p \land q$: T if both p and q are T, else F.
     - $p \lor q$: T if p is T or q is T (or both), else F.
     - $p \to q$: F only if p is T and q is F, else T.
     - $p \leftrightarrow q$: T if p and q have the same truth value, else F.
     - $p \oplus q$: T if p and q have different truth values, else F.
  3. **Tautology, Contradiction, Contingency:**
     - **Tautology:** A compound proposition that is always true, regardless of the truth values of its atomic propositions. Example: $p \lor \neg p$.
     - **Contradiction (or Absurdity):** A compound proposition that is always false. Example: $p \land \neg p$.
     - **Contingency:** A compound proposition that is neither a tautology nor a contradiction (i.e., its truth value depends on the truth values of its atomic propositions).
  4. **Logical Equivalence ($\equiv$):** Two propositions P and Q are logically equivalent if $P \leftrightarrow Q$ is a tautology. This means they always have the same truth value.
  5. **Laws of Logic (Logical Equivalences):**
     - **Identity Laws:**
       - $p \land T \equiv p$
       - $p \lor F \equiv p$

- **Domination Laws:**
  - $p \lor T \equiv T$
  - $p \land F \equiv F$
- **Idempotent Laws:**
  - $p \lor p \equiv p$
  - $p \land p \equiv p$
- **Double Negation Law:** $\neg(\neg p) \equiv p$
- **Commutative Laws:**
  - $p \lor q \equiv q \lor p$
  - $p \land q \equiv q \land p$
- **Associative Laws:**
  - $(p \lor q) \lor r \equiv p \lor (q \lor r)$
  - $(p \land q) \land r \equiv p \land (q \land r)$
- **Distributive Laws:**
  - $p \land (q \lor r) \equiv (p \land q) \lor (p \land r)$
  - $p \lor (q \land r) \equiv (p \lor q) \land (p \lor r)$
- **De Morgan's Laws:**
  - $\neg(p \land q) \equiv \neg p \lor \neg q$
  - $\neg(p \lor q) \equiv \neg p \land \neg q$
- **Absorption Laws:**
  - $p \lor (p \land q) \equiv p$
  - $p \land (p \lor q) \equiv p$
- **Inverse Laws (Complement Laws):**
  - $p \lor \neg p \equiv T$
  - $p \land \neg p \equiv F$
- **Implication Equivalences:**
  - $p \to q \equiv \neg p \lor q$
  - $p \to q \equiv \neg q \to \neg p$ (Contrapositive)
  - $\neg(p \to q) \equiv p \land \neg q$
- **Biconditional Equivalences:**
  - $p \leftrightarrow q \equiv (p \to q) \land (q \to p)$
  - $p \leftrightarrow q \equiv (\neg p \lor q) \land (\neg q \lor p)$
  - $p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$
  - $p \leftrightarrow q \equiv (p \land q) \lor (\neg p \land \neg q)$

6. **Conjunctive Normal Form (CNF) and Disjunctive Normal Form (DNF):**
   - **Literal:** A propositional variable or its negation (e.g., $p, \neg q$).
   - **Clause:** A disjunction of literals (e.g., $p \lor \neg q \lor r$).
   - **CNF:** A conjunction of clauses (e.g., $(p \lor \neg q) \land (r \lor s)$).
   - **Minterm:** A conjunction of literals where each variable appears exactly once (e.g., $p \land \neg q \land r$).
   - **Maxterm:** A disjunction of literals where each variable appears exactly once (e.g., $p \lor \neg q \lor r$).
   - **DNF:** A disjunction of minterms (e.g., $(p \land \neg q) \lor (r \land s)$).
   - Any propositional formula can be converted into an equivalent CNF or DNF.

- **Key Properties and Identities:** All the laws of logic listed above are crucial identities for simplification and proving equivalence. Understanding the truth conditions for each connective is fundamental.
- **Common Pitfalls or Tricky Points:**
  - Confusing implication ($p \to q$) with causation or "if and only if". Remember $p \to q$ is false only when p is true and q is false.
  - Incorrectly applying De Morgan's Laws, especially with more complex expressions.
  - Misinterpreting the biconditional; it implies mutual implication.
  - Errors in constructing truth tables, particularly for propositions with many variables.
  - Assuming logical equivalence based on partial truth values.
- **Standard Problem-Solving Techniques or Shortcuts:**
  - **Truth Tables:** The most direct method for small propositions to check tautology, contradiction, or equivalence.
  - **Logical Equivalences:** Use the laws of logic to simplify expressions or prove equivalence without truth tables. This is often faster for complex expressions.
  - **Substitution:** Replace sub-expressions with their known equivalents.
  - **Proof by Contradiction:** To show P is a tautology, assume $\neg P$ is true and derive a contradiction. To show $P \equiv Q$, assume $\neg(P \leftrightarrow Q)$ is true and derive a contradiction.
  - **Conversion to CNF/DNF:** Useful for standardization and for certain proof methods like resolution.

First Order Logic (Predicate Logic)

First Order Logic (FOL), also known as Predicate Logic, extends propositional logic by allowing us to express more complex statements about objects, their properties, and relationships. It introduces predicates, variables, and quantifiers, enabling reasoning about "all" or "some" elements within a domain.

- **Definition and Core Idea:** FOL allows us to break down propositions into predicates (properties or relations) and arguments (variables or constants). It introduces quantifiers ($\forall$ for "for all" and $\exists$ for "there exists") to make statements about collections of objects, providing a richer expressive power than propositional logic.
- **Important Formulas, Theorems, and Results:**
  1. **Components of FOL:**
     - **Predicates:** Represent properties or relations (e.g., $P(x)$ for "x is prime", $L(x, y)$ for "x loves y").
     - **Variables:** Placeholders for objects (e.g., x, y, z).
     - **Constants:** Specific objects (e.g., "Socrates", "5").
     - **Functions:** Map objects to other objects (e.g., $f(x)$ for "father of x").
     - **Quantifiers:**
       - **Universal Quantifier ($\forall$):** "For all", "for every", "for each". $\forall x P(x)$ means $P(x)$ is true for every x in the domain.
       - **Existential Quantifier ($\exists$):** "There exists", "for some", "for at least one". $\exists x P(x)$ means there is at least one x in the domain for which $P(x)$ is true.
  2. **Scope of Quantifiers, Free and Bound Variables:**
     - The **scope** of a quantifier is the part of the formula to which it applies.
     - A variable is **bound** if it is within the scope of a quantifier using that variable.
     - A variable is **free** if it is not bound by any quantifier.
     - A formula with no free variables is called a **sentence** or **closed formula** and has a definite truth value.
  3. **Quantifier Equivalences (Negation Rules):**
     - $\neg \forall x P(x) \equiv \exists x \neg P(x)$ (It is not true that all x have property P if and only if there exists an x that does not have property P).
     - $\neg \exists x P(x) \equiv \forall x \neg P(x)$ (It is not true that there exists an x with property P if and only if all x do not have property P).
  4. **Distributive Properties of Quantifiers:**
     - $\forall x(P(x) \land Q(x)) \equiv \forall x P(x) \land \forall x Q(x)$
     - $\exists x(P(x) \lor Q(x)) \equiv \exists x P(x) \lor \exists x Q(x)$
     - **Important Non-Equivalences:**
       - $\forall x(P(x) \lor Q(x))$ is NOT equivalent to $\forall x P(x) \lor \forall x Q(x)$. (e.g., "Every number is even or odd" is true, but "Every number is even or every number is odd" is false).
       - $\exists x(P(x) \land Q(x))$ is NOT equivalent to $\exists x P(x) \land \exists x Q(x)$. (e.g., "There is a number that is both even and prime" (2) is true, but "There is an even number and there is a prime number" is also true, but the existence of one doesn't imply the other is the same number).
  5. **Interchange of Quantifiers:**
     - $\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$
     - $\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$
     - **Order Matters for Mixed Quantifiers:**
       - $\forall x \exists y P(x, y)$ (For every x, there exists a y such that P(x,y)) is NOT equivalent to $\exists y \forall x P(x, y)$ (There exists a y such that for every x, P(x,y)).
       - Example: Let $P(x, y)$ be "x loves y". $\forall x \exists y L(x, y)$ means "Everyone loves someone". $\exists y \forall x L(x, y)$ means "There is someone whom everyone loves". These are clearly different.
  6. **Prenex Normal Form (PNF):** A formula is in PNF if all quantifiers are at the beginning of the formula, followed by a quantifier-free formula (called the matrix). Any FOL formula can be converted to an equivalent PNF.
- **Key Properties and Identities:** The quantifier negation rules are fundamental for manipulating and simplifying FOL expressions. Understanding the implications of quantifier order is critical.
- **Common Pitfalls or Tricky Points:**
  - **Incorrect Translation:** Translating natural language statements into FOL is a major source of errors. Pay close attention to words like "all", "some", "no", "only if", "unless".
    - "All A are B": $\forall x(A(x) \to B(x))$ (NOT $\forall x(A(x) \land B(x))$)
    - "Some A are B": $\exists x(A(x) \land B(x))$ (NOT $\exists x(A(x) \to B(x))$)
    - "No A are B": $\forall x(A(x) \to \neg B(x))$ or $\neg \exists x(A(x) \land B(x))$
  - **Scope Errors:** Misplacing parentheses or incorrectly determining the scope of a quantifier.
  - **Variable Binding:** Confusing free and bound variables, or using the same variable name for different purposes in nested scopes.

- **Order of Mixed Quantifiers:** Assuming $\forall x \exists y P(x, y)$ is the same as $\exists y \forall x P(x, y)$.
- **Standard Problem-Solving Techniques or Shortcuts:**
  - **Step-by-step Translation:** Break down complex natural language sentences into smaller parts, translate each, and then combine them. Identify predicates, variables, and connectives first.
  - **Applying Quantifier Equivalences:** Use negation rules to simplify expressions or move negations inwards.
  - **Domain Awareness:** Always consider the domain of discourse when evaluating truth values or translating.
  - **Counterexamples:** To show a statement is false or not equivalent, try to find a specific example in a small domain that makes it false.

Logical Reasoning (Inference)

Logical Reasoning, or Inference, is the process of deriving conclusions from a set of premises using valid rules. It focuses on the structure of arguments to determine whether a conclusion necessarily follows from the premises, regardless of the actual truth of the premises.

- **Definition and Core Idea:** An argument is a sequence of propositions, where the first propositions are premises and the final proposition is the conclusion. An argument is **valid** if, whenever all its premises are true, the conclusion must also be true. The goal is to determine if an argument's structure guarantees the truth of its conclusion given true premises.
- **Important Formulas, Theorems, and Results:**
  1. **Argument Validity:** An argument with premises $P_1, P_2, \ldots, P_n$ and conclusion C is valid if the proposition $(P_1 \wedge P_2 \wedge \ldots \wedge P_n) \rightarrow C$ is a tautology.
  2. **Rules of Inference (for Propositional Logic):** These are fundamental valid argument forms.
     - **Modus Ponens (Law of Detachment):**

$$\begin{array}{l} p \rightarrow q \\ p \\ \hline \therefore q \end{array}$$

       Equivalent to $((p \rightarrow q) \wedge p) \rightarrow q$ being a tautology.
     - **Modus Tollens:**

$$\begin{array}{l} p \rightarrow q \\ \neg q \\ \hline \therefore \neg p \end{array}$$

       Equivalent to $((p \rightarrow q) \wedge \neg q) \rightarrow \neg p$ being a tautology.
     - **Hypothetical Syllogism:**

$$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$$

       Equivalent to $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$ being a tautology.
     - **Disjunctive Syllogism:**

$$\begin{array}{l} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$$

       Equivalent to $((p \vee q) \wedge \neg p) \rightarrow q$ being a tautology.
     - **Addition:**

$$\begin{array}{l} p \\ \hline \therefore p \vee q \end{array}$$

     - **Simplification:**

$$\frac{p \land q}{\therefore p}$$

- **Conjunction:**

$$\begin{array}{l} p \\ q \\ \hline \therefore p \land q \end{array}$$

- **Resolution:**

$$\begin{array}{l} p \lor q \\ \neg q \lor r \\ \hline \therefore p \lor r \end{array}$$

This is particularly useful in automated theorem proving and for converting to CNF.

- **Constructive Dilemma:**

$$\begin{array}{l} (p \to q) \land (r \to s) \\ p \lor r \\ \hline \therefore q \lor s \end{array}$$

- **Destructive Dilemma:**

$$\begin{array}{l} (p \to q) \land (r \to s) \\ \neg q \lor \neg s \\ \hline \therefore \neg p \lor \neg r \end{array}$$

3. **Rules of Inference (for First Order Logic):**
   - **Universal Instantiation (UI):** If $\forall x P(x)$ is true, then $P(c)$ is true for any arbitrary individual c in the domain.

$$\frac{\forall x P(x)}{\therefore P(c)}$$

   - **Universal Generalization (UG):** If $P(c)$ is true for an arbitrary individual c in the domain, then $\forall x P(x)$ is true.

$$\frac{P(c) \text{ for an arbitrary } c}{\therefore \forall x P(x)}$$

   - **Existential Instantiation (EI):** If $\exists x P(x)$ is true, then $P(c)$ is true for some specific individual c in the domain. This c must be a new constant not previously used in the proof.

$$\frac{\exists x P(x)}{\therefore P(c) \text{ (for some new c)}}$$

   - **Existential Generalization (EG):** If $P(c)$ is true for some specific individual c in the domain, then $\exists x P(x)$ is true.

$$\frac{P(c) \text{ for some specific } c}{\therefore \exists x P(x)}$$

4. **Proof Methods:**
   - **Direct Proof:** Assume premises are true, and use inference rules and logical equivalences to derive the conclusion.

    - **Indirect Proof (Proof by Contradiction):** To prove C from premises $P_1, \dots, P_n$, assume $\neg C$ is true along with $P_1, \dots, P_n$, and derive a contradiction (F).
    - **Proof by Contraposition:** To prove $p \to q$, prove its contrapositive $\neg q \to \neg p$.
- **Key Properties and Identities:** The validity of each inference rule is a key property. Understanding that these rules preserve truth is fundamental.
- **Common Pitfalls or Tricky Points:**
  - **Fallacies:** Invalid argument forms that resemble valid ones.
    - **Fallacy of Affirming the Consequent:** $((p \to q) \land q) \to p$ (NOT a tautology). Example: "If it rains, the ground is wet. The ground is wet. Therefore, it rained." (Could be wet from a sprinkler).
    - **Fallacy of Denying the Antecedent:** $((p \to q) \land \neg p) \to \neg q$ (NOT a tautology). Example: "If it rains, the ground is wet. It did not rain. Therefore, the ground is not wet." (Could be wet from a sprinkler).
  - **Confusing Validity with Soundness:** A valid argument can have false premises and a false conclusion. A **sound** argument is a valid argument with all true premises (and thus, a true conclusion). GATE questions usually focus on validity.
  - **Incorrect Application of Quantifier Rules:** Especially with EI and UG, ensure the constant chosen for instantiation is new (for EI) or truly arbitrary (for UG).
  - **Circular Reasoning:** Assuming the conclusion (or something equivalent to it) as a premise.
- **Standard Problem-Solving Techniques or Shortcuts:**
  - **Formal Proofs:** List premises and then systematically apply inference rules and logical equivalences to derive the conclusion, citing the rule for each step.
  - **Truth Tables (for small arguments):** Convert the argument into a single conditional statement $(P_1 \land \dots \land P_n) \to C$ and check if it's a tautology.
  - **Resolution Principle:** Convert all premises and the negation of the conclusion into CNF clauses. Add the negation of the conclusion as a clause. If an empty clause ($\square$) can be derived through resolution, the original argument is valid.
  - **Look for Fallacies:** If an argument resembles a common fallacy, it's likely invalid.

Quick Formula Reference

Propositional Logic Equivalences

- $\neg(\neg p) \equiv p$ (Double Negation)
- $p \lor q \equiv q \lor p$ (Commutative)
- $p \land q \equiv q \land p$ (Commutative)
- $(p \lor q) \lor r \equiv p \lor (q \lor r)$ (Associative)
- $(p \land q) \land r \equiv p \land (q \land r)$ (Associative)
- $p \land (q \lor r) \equiv (p \land q) \lor (p \land r)$ (Distributive)
- $p \lor (q \land r) \equiv (p \lor q) \land (p \lor r)$ (Distributive)
- $\neg(p \land q) \equiv \neg p \lor \neg q$ (De Morgan's)
- $\neg(p \lor q) \equiv \neg p \land \neg q$ (De Morgan's)
- $p \to q \equiv \neg p \lor q$
- $p \to q \equiv \neg q \to \neg p$ (Contrapositive)
- $\neg(p \to q) \equiv p \land \neg q$
- $p \leftrightarrow q \equiv (p \to q) \land (q \to p)$
- $p \leftrightarrow q \equiv (p \land q) \lor (\neg p \land \neg q)$
- $p \lor \neg p \equiv T$ (Inverse Law)
- $p \land \neg p \equiv F$ (Inverse Law)
- $p \lor T \equiv T$ (Domination Law)
- $p \land F \equiv F$ (Domination Law)
- $p \land T \equiv p$ (Identity Law)
- $p \lor F \equiv p$ (Identity Law)
- $p \lor p \equiv p$ (Idempotent Law)
- $p \land p \equiv p$ (Idempotent Law)
- $p \lor (p \land q) \equiv p$ (Absorption Law)
- $p \land (p \lor q) \equiv p$ (Absorption Law)

Quantifier Equivalences

- $\neg \forall x P(x) \equiv \exists x \neg P(x)$
- $\neg \exists x P(x) \equiv \forall x \neg P(x)$

- $\forall x(P(x) \land Q(x)) \equiv \forall xP(x) \land \forall xQ(x)$
- $\exists x(P(x) \lor Q(x)) \equiv \exists xP(x) \lor \exists xQ(x)$
- $\forall x\forall yP(x,y) \equiv \forall y\forall xP(x,y)$
- $\exists x\exists yP(x,y) \equiv \exists y\exists xP(x,y)$

Rules of Inference (Propositional Logic)

- **Modus Ponens:** $(p \to q) \land p \implies q$
- **Modus Tollens:** $(p \to q) \land \neg q \implies \neg p$
- **Hypothetical Syllogism:** $(p \to q) \land (q \to r) \implies p \to r$
- **Disjunctive Syllogism:** $(p \lor q) \land \neg p \implies q$
- **Addition:** $p \implies p \lor q$
- **Simplification:** $p \land q \implies p$
- **Conjunction:** $p, q \implies p \land q$
- **Resolution:** $(p \lor q) \land (\neg q \lor r) \implies p \lor r$

Rules of Inference (First Order Logic)

- **Universal Instantiation (UI):** $\forall xP(x) \implies P(c)$
- **Universal Generalization (UG):** $P(c)$ for arbitrary $c \implies \forall xP(x)$
- **Existential Instantiation (EI):** $\exists xP(x) \implies P(c)$ (for new c)
- **Existential Generalization (EG):** $P(c)$ for specific $c \implies \exists xP(x)$

Important Tips for GATE

1. **Master Truth Tables:** They are the foundation. Be able to quickly construct and interpret truth tables for any compound proposition, especially for 2-3 variables. This helps in verifying tautologies, contradictions, and equivalences.
2. **Memorize Key Equivalences and Inference Rules:** Speed and accuracy in GATE questions heavily rely on your ability to recall and apply De Morgan's laws, implication equivalences, and the common rules of inference (Modus Ponens, Modus Tollens, etc.) without hesitation.
3. **Practice Natural Language Translation:** A significant portion of GATE questions involves translating English sentences into logical expressions (and vice-versa). Pay close attention to keywords like "all", "some", "no", "only if", "unless", and the correct use of quantifiers and connectives.
4. **Understand Quantifier Scope and Order:** Incorrectly interpreting the scope of quantifiers or swapping the order of mixed quantifiers ($\forall x\exists y$ vs. $\exists y\forall x$) is a very common mistake. Always visualize the domain and what each quantifier implies.
5. **Distinguish Validity from Soundness:** GATE questions usually ask about the *validity* of an argument (whether the conclusion logically follows from the premises), not its *soundness* (which also requires premises to be true in the real world). Focus on the logical structure.
6. **Systematic Approach to Proofs:** When asked to prove an argument's validity, don't guess. Start with the premises and systematically apply inference rules and logical equivalences, one step at a time, until you reach the conclusion. Clearly state the rule used for each step.
7. **Identify Common Fallacies:** Be aware of fallacies like "Affirming the Consequent" and "Denying the Antecedent". If an argument structure matches a known fallacy, you can quickly identify it as invalid.
8. **Time Management:** Some logical reasoning problems can be lengthy, especially those involving formal proofs or complex translations. If a question seems too time-consuming, make an educated guess or mark it for review if time permits, and move on to other questions.

3.1 First Order Logic (35)

3.1.1 First Order Logic: GATE CSE 1989 | Question: 14a

Symbolize the expression "Every mother loves her children" in predicate logic.

gate1989 descriptive first-order-logic mathematical-logic

Answer key☟

3.1.2 First Order Logic: GATE CSE 1991 | Question: 15,b

Consider the following first order formula:

$$\begin{pmatrix} \forall x \exists y : R(x,y) \\ \wedge \\ \forall x \forall y : (R(x,y) \implies \neg R(y,x)) \\ \wedge \\ \forall x \forall y \forall z : (R(x,y) \wedge R(y,z) \implies R(x,z)) \\ \wedge \\ \forall x : \neg R(x,x) \end{pmatrix}$$

Does it have finite models?

Is it satisfiable? If so, give a countable model for it.

gate1991 mathematical-logic first-order-logic descriptive

Answer key

3.1.3 First Order Logic: GATE CSE 1992 | Question: 92,xv

Which of the following predicate calculus statements is/are valid?

A. $(\forall(x))P(x) \vee (\forall(x))Q(x) \implies (\forall(x))(P(x) \vee Q(x))$
B. $(\exists(x))P(x) \wedge (\exists(x))Q(x) \implies (\exists(x))(P(x) \wedge Q(x))$
C. $(\forall(x))(P(x) \vee Q(x)) \implies (\forall(x))P(x) \vee (\forall(x))Q(x)$
D. $(\exists(x))(P(x) \vee Q(x)) \implies \sim (\forall(x))P(x) \vee (\exists(x))Q(x)$

gate1992 mathematical-logic normal first-order-logic

Answer key

3.1.4 First Order Logic: GATE CSE 2003 | Question: 32

Which of the following is a valid first order formula? (Here α and β are first order formulae with x as their only free variable)

A. $((\forall x)[\alpha] \Rightarrow (\forall x)[\beta]) \Rightarrow (\forall x)[\alpha \Rightarrow \beta]$
B. $(\forall x)[\alpha] \Rightarrow (\exists x)[\alpha \wedge \beta]$
C. $((\forall x)[\alpha \vee \beta] \Rightarrow (\exists x)[\alpha]) \Rightarrow (\forall x)[\alpha]$
D. $(\forall x)[\alpha \Rightarrow \beta] \Rightarrow (((\forall x)[\alpha]) \Rightarrow (\forall x)[\beta])$

gatecse-2003 mathematical-logic first-order-logic normal

Answer key

3.1.5 First Order Logic: GATE CSE 2003 | Question: 33

Consider the following formula and its two interpretations I_1 and I_2.

$\alpha : (\forall x)\,[P_x \Leftrightarrow (\forall y)\,[Q_{xy} \Leftrightarrow \neg Q_{yy}]] \Rightarrow (\forall x)\,[\neg P_x]$

I_1 : Domain: the set of natural numbers

P_x = 'x is a prime number'

Q_{xy} = 'y divides x'

I_2 : same as I_1 except that P_x = 'x is a composite number'.

Which of the following statements is true?

A. I_1 satisfies α, I_2 does not
B. I_2 satisfies α, I_1 does not
C. Neither I_1 nor I_2 satisfies α
D. Both I_1 and I_2 satisfies α

gatecse-2003 mathematical-logic difficult first-order-logic

Answer key

3.1.6 First Order Logic: GATE CSE 2004 | Question: 23, ISRO2007-32

Identify the correct translation into logical notation of the following assertion.

Some boys in the class are taller than all the girls

Note: $\text{taller}(x, y)$ is true if x is taller than y.

A. $(\exists x)(\text{boy}(x) \rightarrow (\forall y)(\text{girl}(y) \wedge \text{taller}(x, y)))$
B. $(\exists x)(\text{boy}(x) \wedge (\forall y)(\text{girl}(y) \wedge \text{taller}(x, y)))$
C. $(\exists x)(\text{boy}(x) \rightarrow (\forall y)(\text{girl}(y) \rightarrow \text{taller}(x, y)))$
D. $(\exists x)(\text{boy}(x) \wedge (\forall y)(\text{girl}(y) \rightarrow \text{taller}(x, y)))$

gatecse-2004 mathematical-logic easy isro2007 first-order-logic

Answer key

3.1.7 First Order Logic: GATE CSE 2005 | Question: 41

What is the first order predicate calculus statement equivalent to the following?

"Every teacher is liked by some student"

A. $\forall(x)$ $[\text{teacher}\ (x) \rightarrow \exists(y)\ [\text{student}\ (y) \rightarrow \text{likes}\ (y, x)]]$
B. $\forall(x)$ $[\text{teacher}\ (x) \rightarrow \exists(y)\ [\text{student}\ (y) \wedge \text{likes}\ (y, x)]]$
C. $\exists(y)\forall(x)$ $[\text{teacher}\ (x) \rightarrow [\text{student}\ (y) \wedge \text{likes}\ (y, x)]]$
D. $\forall(x)$ $[\text{teacher}\ (x) \wedge \exists(y)\ [\text{student}\ (y) \rightarrow \text{likes}\ (y, x)]]$

gatecse-2005 mathematical-logic easy first-order-logic

Answer key

3.1.8 First Order Logic: GATE CSE 2006 | Question: 26

Which one of the first order predicate calculus statements given below correctly expresses the following English statement?

Tigers and lions attack if they are hungry or threatened.

A. $\forall x[(\text{tiger}(x) \wedge \text{lion}(x)) \rightarrow (\text{hungry}(x) \vee \text{threatened}(x)) \rightarrow \text{attacks}(x)]$
B. $\forall x[(\text{tiger}(x) \vee \text{lion}(x)) \rightarrow (\text{hungry}(x) \vee \text{threatened}(x)) \wedge \text{attacks}(x)]$
C. $\forall x[(\text{tiger}(x) \vee \text{lion}(x)) \rightarrow \text{attacks}(x) \rightarrow (\text{hungry}(x) \vee \text{threatened}(x))]$
D. $\forall x[(\text{tiger}(x) \vee \text{lion}(x)) \rightarrow (\text{hungry}(x) \vee \text{threatened}(x)) \rightarrow \text{attacks}(x)]$

gatecse-2006 mathematical-logic normal first-order-logic tricky

Answer key☟

3.1.9 First Order Logic: GATE CSE 2007 | Question: 22

Let $\text{Graph}(x)$ be a predicate which denotes that x is a graph. Let $\text{Connected}(x)$ be a predicate which denotes that x is connected. Which of the following first order logic sentences **DOES NOT** represent the statement:

"Not every graph is connected"

A. $\neg \forall x \left(\text{Graph}(x) \implies \text{Connected}(x) \right)$

B. $\exists x \left(\text{Graph}(x) \wedge \neg \text{Connected}(x) \right)$

C. $\neg \forall x \left(\neg \text{Graph}(x) \vee \text{Connected}(x) \right)$

D. $\forall x \left(\text{Graph}(x) \implies \neg \text{Connected}(x) \right)$

gatecse-2007 mathematical-logic easy first-order-logic

Answer key☟

3.1.10 First Order Logic: GATE CSE 2008 | Question: 30

Let fsa and pda be two predicates such that $\text{fsa}(x)$ means x is a finite state automaton and $\text{pda}(y)$ means that y is a pushdown automaton. Let equivalent be another predicate such that $\text{equivalent}(a, b)$ means a and b are equivalent. Which of the following first order logic statements represent the following?

Each finite state automaton has an equivalent pushdown automaton

A. $(\forall x \,\text{fsa}\,(x)) \implies (\exists y \,\text{pda}\,(y) \wedge \text{equivalent}\,(x, y))$
B. $\neg \forall y \,(\exists x \,\text{fsa}\,(x) \implies \text{pda}\,(y) \wedge \text{equivalent}\,(x, y))$
C. $\forall x \exists y \,(\text{fsa}\,(x) \wedge \text{pda}\,(y) \wedge \text{equivalent}\,(x, y))$
D. $\forall x \exists y \,(\text{fsa}\,(y) \wedge \text{pda}\,(x) \wedge \text{equivalent}\,(x, y))$

gatecse-2008 easy mathematical-logic first-order-logic

Answer key☟

3.1.11 First Order Logic: GATE CSE 2009 | Question: 23

Which one of the following is the most appropriate logical formula to represent the statement?

"Gold and silver ornaments are precious".

The following notations are used:

- $G(x)$: x is a gold ornament
- $S(x)$: x is a silver ornament
- $P(x)$: x is precious

A. $\forall x (P(x) \implies (G(x) \wedge S(x)))$

B. $\forall x ((G(x) \wedge S(x)) \implies P(x))$

C. $\exists x ((G(x) \wedge S(x)) \implies P(x))$

D. $\forall x ((G(x) \vee S(x)) \implies P(x))$

gatecse-2009 mathematical-logic easy first-order-logic

Answer key☟

3.1.12 First Order Logic: GATE CSE 2009 | Question: 26

Consider the following well-formed formulae:

I. $\neg \forall x (P(x))$
II. $\neg \exists x (P(x))$
III. $\neg \exists x (\neg P(x))$
IV. $\exists x (\neg P(x))$

Which of the above are equivalent?

A. I and III

B. I and IV

C. II and III

D. II and IV

gatecse-2009 mathematical-logic normal first-order-logic

Answer key

3.1.13 First Order Logic: GATE CSE 2010 | Question: 30

Suppose the predicate $F(x, y, t)$ is used to represent the statement that person x can fool person y at time t.

Which one of the statements below expresses best the meaning of the formula,

$$\forall x \exists y \exists t (\neg F(x, y, t))$$

A. Everyone can fool some person at some time
B. No one can fool everyone all the time
C. Everyone cannot fool some person all the time
D. No one can fool some person at some time

gatecse-2010 mathematical-logic easy first-order-logic

Answer key

3.1.14 First Order Logic: GATE CSE 2011 | Question: 30

Which one of the following options is CORRECT given three positive integers x, y and z, and a predicate

$$P(x) = \neg (x = 1) \wedge \forall y (\exists z (x = y * z) \Rightarrow (y = x) \vee (y = 1))$$

A. $P(x)$ being true means that x is a prime number
B. $P(x)$ being true means that x is a number other than 1
C. $P(x)$ is always true irrespective of the value of x
D. $P(x)$ being true means that x has exactly two factors other than 1 and x

gatecse-2011 mathematical-logic normal first-order-logic

Answer key

3.1.15 First Order Logic: GATE CSE 2012 | Question: 13

What is the correct translation of the following statement into mathematical logic?

“Some real numbers are rational”

A. $\exists x(\text{real}(x) \vee \text{rational}(x))$
B. $\forall x(\text{real}(x) \rightarrow \text{rational}(x))$
C. $\exists x(\text{real}(x) \wedge \text{rational}(x))$
D. $\exists x(\text{rational}(x) \rightarrow \text{real}(x))$

gatecse-2012 mathematical-logic easy first-order-logic

Answer key

3.1.16 First Order Logic: GATE CSE 2013 | Question: 27

What is the logical translation of the following statement?

"None of my friends are perfect."

A. $\exists x(F(x) \wedge \neg P(x))$
B. $\exists x(\neg F(x) \wedge P(x))$
C. $\exists x(\neg F(x) \wedge \neg P(x))$
D. $\neg \exists x(F(x) \wedge P(x))$

gatecse-2013 mathematical-logic easy first-order-logic

Answer key

3.1.17 First Order Logic: GATE CSE 2013 | Question: 47

Which one of the following is **NOT** logically equivalent to $\neg \exists x(\forall y(\alpha) \wedge \forall z(\beta))$?

A. $\forall x(\exists z(\neg \beta) \rightarrow \forall y(\alpha))$
B. $\forall x(\forall z(\beta) \rightarrow \exists y(\neg \alpha))$
C. $\forall x(\forall y(\alpha) \rightarrow \exists z(\neg \beta))$
D. $\forall x(\exists y(\neg \alpha) \rightarrow \exists z(\neg \beta))$

mathematical-logic normal marks-to-all gatecse-2013 first-order-logic

Answer key

3.1.18 First Order Logic: GATE CSE 2014 | Set 1 | Question: 1

Consider the statement

"Not all that glitters is gold”

Predicate glitters(x) is true if x glitters and predicate gold(x) is true if x is gold. Which one of the following logical formulae represents the above statement?

A. $\forall x : \text{glitters}(x) \Rightarrow \neg\text{gold}(x)$
B. $\forall x : \text{gold}(x) \Rightarrow \text{glitters}(x)$
C. $\exists x : \text{gold}(x) \wedge \neg\text{glitters}(x)$
D. $\exists x : \text{glitters}(x) \wedge \neg\text{gold}(x)$

gatecse-2014-set1 mathematical-logic first-order-logic

Answer key

3.1.19 First Order Logic: GATE CSE 2014 | Set 3 | Question: 53

The CORRECT formula for the sentence, "not all Rainy days are Cold" is

A. $\forall d(\text{Rainy}(d) \wedge \sim\text{Cold}(d))$
B. $\forall d(\sim\text{Rainy}(d) \rightarrow \text{Cold}(d))$
C. $\exists d(\sim\text{Rainy}(d) \rightarrow \text{Cold}(d))$
D. $\exists d(\text{Rainy}(d) \wedge \sim\text{Cold}(d))$

gatecse-2014-set3 mathematical-logic easy first-order-logic

Answer key

3.1.20 First Order Logic: GATE CSE 2015 | Set 2 | Question: 55

Which one of the following well-formed formulae is a tautology?

A. $\forall x\, \exists y\, R(x,y) \leftrightarrow \exists y\, \forall x\, R(x,y)$
B. $(\forall x\, [\exists y\, R(x,y) \rightarrow S(x,y)]) \rightarrow \forall x\, \exists y\, S(x,y)$
C. $[\forall x\, \exists y\, (P(x,y) \rightarrow R(x,y))] \leftrightarrow [\forall x\, \exists y\, (\neg P(x,y) \vee R(x,y))]$
D. $\forall x\, \forall y\, P(x,y) \rightarrow \forall x\, \forall y\, P(y,x)$

gatecse-2015-set2 mathematical-logic normal first-order-logic

Answer key

3.1.21 First Order Logic: GATE CSE 2016 | Set 2 | Question: 27

Which one of the following well-formed formulae in predicate calculus is **NOT** valid ?

A. $(\forall_x p(x) \implies \forall_x q(x)) \implies (\exists_x \neg p(x) \vee \forall_x q(x))$
B. $(\exists x p(x) \vee \exists x q(x)) \implies \exists x (p(x) \vee q(x))$
C. $\exists x (p(x) \wedge q(x)) \implies (\exists x p(x) \wedge \exists x q(x))$
D. $\forall x (p(x) \vee q(x)) \implies (\forall x p(x) \vee \forall x q(x))$

gatecse-2016-set2 mathematical-logic first-order-logic normal

Answer key

3.1.22 First Order Logic: GATE CSE 2017 | Set 1 | Question: 02

Consider the first-order logic sentence $F : \forall x(\exists y R(x,y))$. Assuming non-empty logical domains, which of the sentences below are *implied* by F?

I. $\exists y(\exists x R(x,y))$
II. $\exists y(\forall x R(x,y))$
III. $\forall y(\exists x R(x,y))$
IV. $\neg\exists x(\forall y \neg R(x,y))$

A. IV only
B. I and IV only
C. II only
D. II and III only

gatecse-2017-set1 mathematical-logic first-order-logic

Answer key

3.1.23 First Order Logic: GATE CSE 2018 | Question: 28

Consider the first-order logic sentence

$$\varphi \equiv \exists s \exists t \exists u \forall v \forall w \forall x \forall y \, \psi(s,t,u,v,w,x,y)$$

where $\psi(s,t,u,v,w,x,y,)$ is a quantifier-free first-order logic formula using only predicate symbols, and possibly equality, but no function symbols. Suppose φ has a model with a universe containing 7 elements.

Which one of the following statements is necessarily true?

A. There exists at least one model of φ with universe of size less than or equal to 3
B. There exists no model of φ with universe of size less than or equal to 3
C. There exists no model of φ with universe size of greater than 7
D. Every model of φ has a universe of size equal to 7

gatecse-2018 mathematical-logic normal first-order-logic two-marks

Answer key

3.1.24 First Order Logic: GATE CSE 2019 | Question: 35

Consider the first order predicate formula φ:

$\forall x[(\forall z \, z|x \Rightarrow ((z=x) \vee (z=1))) \rightarrow \exists w(w>x) \wedge (\forall z \, z|w \Rightarrow ((w=z) \vee (z=1)))]$

Here $a \mid b$ denotes that $'a \text{ divides } b'$, where a and b are integers. Consider the following sets:

- $S_1 : \{1,2,3,\dots,100\}$
- S_2 : Set of all positive integers
- S_3 : Set of all integers

Which of the above sets satisfy φ?

A. S_1 and S_2
B. S_1 and S_3
C. S_2 and S_3
D. S_1, S_2 and S_3

gatecse-2019 discrete-mathematics mathematical-logic first-order-logic two-marks normal

Answer key

3.1.25 First Order Logic: GATE CSE 2020 | Question: 39

Which one of the following predicate formulae is NOT logically valid?

Note that W is a predicate formula without any free occurrence of x.

A. $\forall x(p(x) \vee W) \equiv \forall x \, p(x) \vee W$
B. $\exists x(p(x) \wedge W) \equiv \exists x \, p(x) \wedge W$
C. $\forall x(p(x) \rightarrow W) \equiv \forall x \, p(x) \rightarrow W$
D. $\exists x(p(x) \rightarrow W) \equiv \forall x \, p(x) \rightarrow W$

gatecse-2020 first-order-logic mathematical-logic two-marks normal

Answer key

3.1.26 First Order Logic: GATE CSE 2023 | Question: 16

Geetha has a conjecture about integers, which is of the form

$$\forall x(P(x) \implies \exists y Q(x,y)),$$

where P is a statement about integers, and Q is a statement about pairs of integers. Which of the following (one or more) option(s) would *imply* Geetha's conjecture?

A. $\exists x(P(x) \wedge \forall y Q(x, y))$
B. $\forall x \forall y Q(x, y)$
C. $\exists y \forall x(P(x) \Longrightarrow Q(x, y))$
D. $\exists x(P(x) \wedge \exists y Q(x, y))$

gatecse-2023 mathematical-logic first-order-logic multiple-selects one-mark difficult

Answer key

3.1.27 First Order Logic: GATE CSE 2025 | Set 1 | Question: 38

Which of the following predicate logic formulae/formula is/are CORRECT representation(s) of the statement: "Everyone has exactly one mother"?

The meanings of the predicates used are:

- mother (y, x) : y is the mother of x
- noteq (x, y) : x and y are not equal

A. $\forall x \exists y \exists z(\text{mother}(y, x) \wedge \neg \text{mother}(z, x))$
B. $\forall x \exists y[\text{mother}(y, x) \wedge \forall z(\text{noteq}(z, y) \rightarrow \neg \text{mother}(z, x))]$
C. $\forall x \forall y[\text{mother}(y, x) \rightarrow \exists z(\text{mother}(z, x) \wedge \neg \text{note} q(z, y))]$
D. $\forall x \exists y[\text{mother}(y, x) \wedge \neg \exists z(\text{note} q(z, y) \wedge \text{mother}(z, x))]$

gatecse2025-set1 mathematical-logic first-order-logic multiple-selects two-marks

Answer key

3.1.28 First Order Logic: GATE CSE 2025 | Set 2 | Question: 5

Let $P(x)$ be an arbitrary predicate over the domain of natural numbers. Which ONE of the following statements is TRUE?

A. $(P(0) \wedge (\forall x[P(x) \Rightarrow P(x+1)])) \Rightarrow \forall x P(x))$
B. $(P(0) \wedge (\forall x[P(x) \Rightarrow P(x-1)])) \Rightarrow \forall x P(x))$
C. $(P(1000) \wedge (\forall x[P(x) \Rightarrow P(x-1)])) \Rightarrow \forall x P(x))$
D. $(P(1000) \wedge (\forall x[P(x) \Rightarrow P(x+1)])) \Rightarrow \forall x P(x))$

gatecse2025-set2 mathematical-logic first-order-logic one-mark

Answer key

3.1.29 First Order Logic: GATE CSE 2026 | Set 2 | Question: 1

For two different persons x and y, the predicate $M(x, y)$ denotes that x knows y. Consider the following statement.

There is a person who does not know anyone else, but that person is known by everyone else.

Which one of the following expressions represents the above statement?

A. $(\exists y)(\forall x)((x \neq y) \rightarrow (M(x, y) \wedge \neg M(y, x)))$
B. $(\forall y)(\exists x)((x \neq y) \rightarrow (M(x, y) \wedge \neg M(y, x)))$
C. $(\exists y)(\exists x)((x \neq y) \rightarrow (M(x, y) \wedge \neg M(y, x)))$
D. $(\forall y)(\forall x)((x \neq y) \rightarrow (M(x, y) \wedge \neg M(y, x)))$

gatecse-2026-set2 mathematical-logic first-order-logic one-mark

Answer key

3.1.30 First Order Logic: GATE IT 2004 | Question: 3

Let $a(x,y), b(x,y,)$ and $c(x,y)$ be three statements with variables x and y chosen from some universe. Consider the following statement:

$$(\exists x)(\forall y)[(a(x,y) \wedge b(x,y)) \wedge \neg c(x,y)]$$

Which one of the following is its equivalent?

A. $(\forall x)(\exists y)[(a(x,y) \vee b(x,y)) \rightarrow c(x,y)]$
B. $(\exists x)(\forall y)[(a(x,y) \vee b(x,y)) \wedge \neg c(x,y)]$
C. $\neg(\forall x)(\exists y)[(a(x,y) \wedge b(x,y)) \rightarrow c(x,y)]$
D. $\neg(\forall x)(\exists y)[(a(x,y) \vee b(x,y)) \rightarrow c(x,y)]$

gateit-2004 mathematical-logic normal discrete-mathematics first-order-logic

Answer key

3.1.31 First Order Logic: GATE IT 2005 | Question: 36

Let $P(x)$ and $Q(x)$ be arbitrary predicates. Which of the following statements is always **TRUE**?

A. $((\forall x\,(P\,(x) \vee Q\,(x)))) \implies ((\forall x P\,(x)) \vee (\forall x Q\,(x)))$
B. $(\forall x\,(P\,(x) \implies Q\,(x))) \implies ((\forall x P\,(x)) \implies (\forall x Q\,(x)))$
C. $(\forall x\,(P\,(x)) \implies \forall x\,(Q\,(x))) \implies (\forall x\,(P\,(x) \implies Q\,(x)))$
D. $(\forall x\,(P\,(x)) \Leftrightarrow (\forall x\,(Q\,(x)))) \implies (\forall x\,(P\,(x) \Leftrightarrow Q\,(x)))$

gateit-2005 mathematical-logic first-order-logic normal

Answer key

3.1.32 First Order Logic: GATE IT 2006 | Question: 21

Consider the following first order logic formula in which R is a binary relation symbol.

$$\forall x \forall y (R(x,y) \implies R(y,x))$$

The formula is

A. satisfiable and valid
B. satisfiable and so is its negation
C. unsatisfiable but its negation is valid
D. satisfiable but its negation is unsatisfiable

gateit-2006 mathematical-logic normal first-order-logic

Answer key

3.1.33 First Order Logic: GATE IT 2007 | Question: 21

Which one of these first-order logic formulae is valid?

A. $\forall x\,(P\,(x) \implies Q\,(x)) \implies (\forall x P\,(x) \implies \forall x Q\,(x))$
B. $\exists x\,(P\,(x) \vee Q\,(x)) \implies (\exists x P\,(x) \implies \exists x Q\,(x))$
C. $\exists x\,(P\,(x) \wedge Q\,(x)) \iff (\exists x P\,(x) \wedge \exists x Q\,(x))$
D. $\forall x \exists y P\,(x,y) \implies \exists y \forall x P\,(x,y)$

gateit-2007 mathematical-logic normal first-order-logic

Answer key

3.1.34 First Order Logic: GATE IT 2008 | Question: 21

Which of the following first order formulae is logically valid? Here $\alpha(x)$ is a first order formula with x as a free variable, and β is a first order formula with no free variable.

A. $[\beta \rightarrow (\exists x, \alpha(x))] \rightarrow [\forall x, \beta \rightarrow \alpha(x)]$
B. $[\exists x, \beta \rightarrow \alpha(x)] \rightarrow [\beta \rightarrow (\forall x, \alpha(x))]$
C. $[(\exists x, \alpha(x)) \rightarrow \beta] \rightarrow [\forall x, \alpha(x) \rightarrow \beta]$
D. $[(\forall x, \alpha(x)) \rightarrow \beta] \rightarrow [\forall x, \alpha(x) \rightarrow \beta]$

gateit-2008 first-order-logic normal

Answer key

3.1.35 First Order Logic: GATE IT 2008 | Question: 22

Which of the following is the negation of $[\forall x, \alpha \rightarrow (\exists y, \beta \rightarrow (\forall u, \exists v, y))]$

A. $[\exists x, \alpha \rightarrow (\forall y, \beta \rightarrow (\exists u, \forall v, y))]$
B. $[\exists x, \alpha \rightarrow (\forall y, \beta \rightarrow (\exists u, \forall v, \neg y))]$
C. $[\forall x, \neg\alpha \rightarrow (\exists y, \neg\beta \rightarrow (\forall u, \exists v, \neg y))]$
D. $[\exists x, \alpha \wedge (\forall y, \beta \wedge (\exists u, \forall v, \neg y))]$

gateit-2008 mathematical-logic normal first-order-logic

Answer key

3.2 Logical Reasoning (3)

3.2.1 Logical Reasoning: GATE CSE 2012 | Question: 1

Consider the following logical inferences.

I_1: If it rains then the cricket match will not be played.
The cricket match was played.
Inference: There was no rain.

I_2: If it rains then the cricket match will not be played.
It did not rain.
Inference: The cricket match was played.

Which of the following is **TRUE**?

A. Both I_1 and I_2 are correct inferences
B. I_1 is correct but I_2 is not a correct inference
C. I_1 is not correct but I_2 is a correct inference
D. Both I_1 and I_2 are not correct inferences

gatecse-2012 mathematical-logic easy logical-reasoning

Answer key

3.2.2 Logical Reasoning: GATE CSE 2015 | Set 2 | Question: 3

Consider the following two statements.

- S_1: If a candidate is known to be corrupt, then he will not be elected
- S_2: If a candidate is kind, he will be elected

Which one of the following statements follows from S_1 and S_2 as per sound inference rules of logic?

A. If a person is known to be corrupt, he is kind
B. If a person is not known to be corrupt, he is not kind
C. If a person is kind, he is not known to be corrupt
D. If a person is not kind, he is not known to be corrupt

gatecse-2015-set2 mathematical-logic normal logical-reasoning

Answer key

3.2.3 Logical Reasoning: GATE CSE 2015 | Set 3 | Question: 24

In a room there are only two types of people, namely Type 1 and Type 2. Type 1 people always tell the truth and Type 2 people always lie. You give a fair coin to a person in that room, without knowing which type he is from and tell him to toss it and hide the result from you till you ask for it. Upon asking the person replies the following

"The result of the toss is head if and only if I am telling the truth"

Which of the following options is correct?

A. The result is head
B. The result is tail
C. If the person is of Type 2, then the result is tail
D. If the person is of Type 1, then the result is tail

gatecse-2015-set3 mathematical-logic difficult logical-reasoning

Answer key

3.3 Propositional Logic (40)

3.3.1 Propositional Logic: GATE CSE 1987 | Question: 10e

Show that the conclusion $(r \to q)$ follows from the premises: $p, (p \to q) \lor (p \land (r \to q))$

gate1987 mathematical-logic propositional-logic proof descriptive

Answer key

3.3.2 Propositional Logic: GATE CSE 1988 | Question: 2vii

Define the validity of a well-formed formula(wff)?

gate1988 descriptive mathematical-logic propositional-logic

Answer key

3.3.3 Propositional Logic: GATE CSE 1989 | Question: 3-v

Which of the following well-formed formulas are equivalent?

A. $P \to Q$
B. $\neg Q \to \neg P$
C. $\neg P \lor Q$
D. $\neg Q \to P$

gate1989 normal mathematical-logic propositional-logic multiple-selects

Answer key

3.3.4 Propositional Logic: GATE CSE 1990 | Question: 3-x

Indicate which of the following well-formed formulae are valid:

A. $(P \Rightarrow Q) \land (Q \Rightarrow R) \Rightarrow (P \Rightarrow R)$
B. $(P \Rightarrow Q) \Rightarrow (\neg P \Rightarrow \neg Q)$
C. $(P \land (\neg P \lor \neg Q)) \Rightarrow Q$
D. $(P \Rightarrow R) \lor (Q \Rightarrow R) \Rightarrow ((P \lor Q) \Rightarrow R)$

gate1990 normal mathematical-logic propositional-logic multiple-selects

Answer key

3.3.5 Propositional Logic: GATE CSE 1991 | Question: 03,xii

If F_1, F_2 and F_3 are propositional formulae such that $F_1 \land F_2 \to F_3$ and $F_1 \land F_2 \to \sim F_3$ are both tautologies, then which of the following is true:

A. Both F_1 and F_2 are tautologies
B. The conjunction $F_1 \land F_2$ is not satisfiable
C. Neither is tautologous
D. Neither is satisfiable
E. None of the above

gate1991 mathematical-logic normal propositional-logic multiple-selects

Answer key☟

3.3.6 Propositional Logic: GATE CSE 1992 | Question: 02,xvi

Which of the following is/are a tautology?

A. $a \vee b \to b \wedge c$
B. $a \wedge b \to b \vee c$
C. $a \vee b \to (b \to c)$
D. $a \to b \to (b \to c)$

gate1992 mathematical-logic easy propositional-logic

Answer key☟

3.3.7 Propositional Logic: GATE CSE 1992 | Question: 15.a

Use Modus ponens $(A, A \to B| = B)$ or resolution to show that the following set is inconsistent:

1. $Q(x) \to P(x) \vee \sim R(a)$
2. $R(a) \vee \sim Q(a)$
3. $Q(a)$
4. $\sim P(y)$

where x and y are universally quantified variables, a is a constant and P, Q, R are monadic predicates.

gate1992 normal mathematical-logic propositional-logic descriptive

Answer key☟

3.3.8 Propositional Logic: GATE CSE 1993 | Question: 18

Show that proposition C is a logical consequence of the formula

$$A \wedge (A \to (B \vee C)) \wedge (B \to \neg A)$$

using truth tables.

gate1993 mathematical-logic normal propositional-logic proof descriptive

Answer key☟

3.3.9 Propositional Logic: GATE CSE 1993 | Question: 8.2

The proposition $p \wedge (\sim p \vee q)$ is:

A. a tautology
B. logically equivalent to $p \wedge q$
C. logically equivalent to $p \vee q$
D. a contradiction
E. none of the above

gate1993 mathematical-logic easy propositional-logic

Answer key☟

3.3.10 Propositional Logic: GATE CSE 1994 | Question: 3.13

Let p and q be propositions. Using only the Truth Table, decide whether

- $p \iff q$ does not imply $p \to \neg q$

is **True** or **False**.

gate1994 mathematical-logic normal propositional-logic true-false

Answer key☟

3.3.11 Propositional Logic: GATE CSE 1995 | Question: 13

Obtain the principal (canonical) conjunctive normal form of the propositional formula

$$(p \wedge q) \vee (\neg q \wedge r)$$

where $\wedge$ is logical and, $\vee$ is inclusive or and $\neg$ is negation.

gate1995 mathematical-logic propositional-logic normal descriptive

Answer key

3.3.12 Propositional Logic: GATE CSE 1995 | Question: 2.19

If the proposition $\neg p \rightarrow q$ is true, then the truth value of the proposition $\neg p \vee (p \rightarrow q)$, where $\neg$ is negation, $\vee$ is inclusive OR and $\rightarrow$ is implication, is

A. True
B. Multiple Values
C. False
D. Cannot be determined

gate1995 mathematical-logic normal propositional-logic

Answer key

3.3.13 Propositional Logic: GATE CSE 1996 | Question: 2.3

Which of the following is NOT True?
(Read $\wedge$ as AND, $\vee$ as OR, $\neg$ as NOT, $\rightarrow$ as one way implication and $\leftrightarrow$ as two way implication)

A. $((x \rightarrow y) \wedge x) \rightarrow y$
B. $((\neg x \rightarrow y) \wedge (\neg x \rightarrow \neg y)) \rightarrow x$
C. $(x \rightarrow (x \vee y))$
D. $((x \vee y) \leftrightarrow (\neg x \rightarrow \neg y))$

gate1996 mathematical-logic normal propositional-logic

Answer key

3.3.14 Propositional Logic: GATE CSE 1997 | Question: 3.2

Which of the following propositions is a tautology?

A. $(p \vee q) \rightarrow p$
B. $p \vee (q \rightarrow p)$
C. $p \vee (p \rightarrow q)$
D. $p \rightarrow (p \rightarrow q)$

gate1997 mathematical-logic easy propositional-logic

Answer key

3.3.15 Propositional Logic: GATE CSE 1998 | Question: 1.5

What is the converse of the following assertion?

- I stay only if you go

A. I stay if you go
B. If I stay then you go
C. If you do not go then I do not stay
D. If I do not stay then you go

gate1998 mathematical-logic easy propositional-logic

Answer key

3.3.16 Propositional Logic: GATE CSE 1999 | Question: 14

Show that the formula $[(\sim p \vee q) \Rightarrow (q \Rightarrow p)]$ is not a tautology.

Let A be a tautology and B any other formula. Prove that $(A \vee B)$ is a tautology.

gate1999 mathematical-logic normal propositional-logic proof descriptive

Answer key

3.3.17 Propositional Logic: GATE CSE 2000 | Question: 2.7

Let a, b, c, d be propositions. Assume that the equivalence $a \Leftrightarrow (b \vee \neg b)$ and $b \Leftrightarrow c$ hold. Then the truth-value of the formula $(a \wedge b) \rightarrow (a \wedge c) \vee d$ is always

A. True
B. False
C. Same as the truth-value of b
D. Same as the truth-value of d

gatecse-2000 mathematical-logic normal propositional-logic

Answer key

3.3.18 Propositional Logic: GATE CSE 2001 | Question: 1.3

Consider two well-formed formulas in propositional logic

$$F_1 : P \Rightarrow \neg P \qquad F_2 : (P \Rightarrow \neg P) \vee (\neg P \Rightarrow P)$$

Which one of the following statements is correct?

A. F_1 is satisfiable, F_2 is valid
B. F_1 unsatisfiable, F_2 is satisfiable
C. F_1 is unsatisfiable, F_2 is valid
D. F_1 and F_2 are both satisfiable

gatecse-2001 mathematical-logic easy propositional-logic

Answer key

3.3.19 Propositional Logic: GATE CSE 2002 | Question: 1.8

"If X then Y unless Z" is represented by which of the following formulas in propositional logic? ("$\neg$" is negation, "$\wedge$" is conjunction, and "$\rightarrow$" is implication)

A. $(X \wedge \neg Z) \rightarrow Y$
B. $(X \wedge Y) \rightarrow \neg Z$
C. $X \rightarrow (Y \wedge \neg Z)$
D. $(X \rightarrow Y) \wedge \neg Z$

gatecse-2002 mathematical-logic normal propositional-logic

Answer key

3.3.20 Propositional Logic: GATE CSE 2002 | Question: 5b

Determine whether each of the following is a tautology, a contradiction, or neither ("$\vee$" is disjunction, "$\wedge$" is conjunction, "$\rightarrow$" is implication, "$\neg$" is negation, and "$\leftrightarrow$" is biconditional (if and only if).

1. $A \leftrightarrow (A \vee A)$
2. $(A \vee B) \rightarrow B$
3. $A \wedge (\neg(A \vee B))$

gatecse-2002 mathematical-logic easy descriptive propositional-logic

Answer key

3.3.21 Propositional Logic: GATE CSE 2003 | Question: 72

The following resolution rule is used in logic programming.
Derive clause $(P \vee Q)$ from clauses $(P \vee R), (Q \vee \neg R)$
Which of the following statements related to this rule is FALSE?

A. $((P \vee R) \wedge (Q \vee \neg R)) \Rightarrow (P \vee Q)$ is logically valid
B. $(P \vee Q) \Rightarrow ((P \vee R) \wedge (Q \vee \neg R))$ is logically valid
C. $(P \vee Q)$ is satisfiable if and only if $(P \vee R) \wedge (Q \vee \neg R)$ is satisfiable
D. $(P \vee Q) \Rightarrow \text{FALSE}$ if and only if both P and Q are unsatisfiable

gatecse-2003 mathematical-logic normal propositional-logic

Answer key

3.3.22 Propositional Logic: GATE CSE 2004 | Question: 70

The following propositional statement is $(P \implies (Q \vee R)) \implies ((P \wedge Q) \implies R)$

A. satisfiable but not valid
B. valid
C. a contradiction
D. None of the above

gatecse-2004 mathematical-logic normal propositional-logic

Answer key

3.3.23 Propositional Logic: GATE CSE 2005 | Question: 40

Let $P, Q,$ and R be three atomic propositional assertions. Let X denote $(P \vee Q) \to R$ and Y denote $(P \to R) \vee (Q \to R)$. Which one of the following is a tautology?

A. $X \equiv Y$
B. $X \to Y$
C. $Y \to X$
D. $\neg Y \to X$

gatecse-2005 mathematical-logic propositional-logic normal

Answer key

3.3.24 Propositional Logic: GATE CSE 2006 | Question: 27

Consider the following propositional statements:

- $P_1 : ((A \wedge B) \to C)) \equiv ((A \to C) \wedge (B \to C))$
- $P_2 : ((A \vee B) \to C)) \equiv ((A \to C) \vee (B \to C))$

Which one of the following is true?

A. P_1 is a tautology, but not P_2
B. P_2 is a tautology, but not P_1
C. P_1 and P_2 are both tautologies
D. Both P_1 and P_2 are not tautologies

gatecse-2006 mathematical-logic normal propositional-logic

Answer key

3.3.25 Propositional Logic: GATE CSE 2008 | Question: 31

P and Q are two propositions. Which of the following logical expressions are equivalent?

I. $P \vee \neg Q$
II. $\neg(\neg P \wedge Q)$
III. $(P \wedge Q) \vee (P \wedge \neg Q) \vee (\neg P \wedge \neg Q)$
IV. $(P \wedge Q) \vee (P \wedge \neg Q) \vee (\neg P \wedge Q)$

A. Only I and II
B. Only I, II and III
C. Only I, II and IV
D. All of I, II, III and IV

gatecse-2008 normal mathematical-logic propositional-logic

Answer key

3.3.26 Propositional Logic: GATE CSE 2009 | Question: 24

The binary operation $\square$ is defined as follows

P	Q	P$\square$Q
T	T	T
T	F	T
F	T	F
F	F	T

Which one of the following is equivalent to $P \vee Q$?

A. $\neg Q \square \neg P$
B. $P \square \neg Q$
C. $\neg P \square Q$
D. $\neg P \square \neg Q$

gatecse-2009 mathematical-logic easy propositional-logic

Answer key

3.3.27 Propositional Logic: GATE CSE 2014 | Set 1 | Question: 53

Which one of the following propositional logic formulas is TRUE when exactly two of p, q and r are TRUE?

A. $((p \leftrightarrow q) \wedge r) \vee (p \wedge q \wedge \sim r)$
B. $(\sim (p \leftrightarrow q) \wedge r) \vee (p \wedge q \wedge \sim r)$
C. $((p \rightarrow q) \wedge r) \vee (p \wedge q \wedge \sim r)$
D. $(\sim (p \leftrightarrow q) \wedge r) \wedge (p \wedge q \wedge \sim r)$

gatecse-2014-set1 mathematical-logic normal propositional-logic

Answer key

3.3.28 Propositional Logic: GATE CSE 2014 | Set 2 | Question: 53

Which one of the following Boolean expressions is NOT a tautology?

A. $((a \rightarrow b) \wedge (b \rightarrow c)) \rightarrow (a \rightarrow c)$
B. $(a \rightarrow c) \rightarrow (\sim b \rightarrow (a \wedge c))$
C. $(a \wedge b \wedge c) \rightarrow (c \vee a)$
D. $a \rightarrow (b \rightarrow a)$

gatecse-2014-set2 mathematical-logic propositional-logic normal

Answer key

3.3.29 Propositional Logic: GATE CSE 2014 | Set 3 | Question: 1

Consider the following statements:

- P: Good mobile phones are not cheap
- Q: Cheap mobile phones are not good

L: P implies Q
M: Q implies P
N: P is equivalent to Q

Which one of the following about L, M, and N is CORRECT?

A. Only L is TRUE.
B. Only M is TRUE.
C. Only N is TRUE.
D. L, M and N are TRUE.

gatecse-2014-set3 mathematical-logic easy propositional-logic

Answer key

3.3.30 Propositional Logic: GATE CSE 2015 | Set 1 | Question: 14

Which one of the following is NOT equivalent to $p \leftrightarrow q$?

A. $(\neg p \vee q) \wedge (p \vee \neg q)$
B. $(\neg p \vee q) \wedge (q \rightarrow p)$
C. $(\neg p \wedge q) \vee (p \wedge \neg q)$
D. $(\neg p \wedge \neg q) \vee (p \wedge q)$

gatecse-2015-set1 mathematical-logic easy propositional-logic

Answer key

3.3.31 Propositional Logic: GATE CSE 2016 | Set 1 | Question: 1

Let p, q, r, s represents the following propositions.

- $p: x \in \{8, 9, 10, 11, 12\}$
- $q: x$ is a composite number.
- $r: x$ is a perfect square.
- $s: x$ is a prime number.

The integer $x \geq 2$ which satisfies $\neg((p \Rightarrow q) \wedge (\neg r \vee \neg s))$ is ________.

gatecse-2016-set1 mathematical-logic normal numerical-answers propositional-logic

Answer key☟

3.3.32 Propositional Logic: GATE CSE 2016 | Set 2 | Question: 01

Consider the following expressions:

i. $false$
ii. Q
iii. $true$
iv. $P \vee Q$
v. $\neg Q \vee P$

The number of expressions given above that are logically implied by $P \wedge (P \Rightarrow Q)$ is ________.

gatecse-2016-set2 mathematical-logic normal numerical-answers propositional-logic

Answer key☟

3.3.33 Propositional Logic: GATE CSE 2017 | Set 1 | Question: 01

The statement $(\neg p) \Rightarrow (\neg q)$ is logically equivalent to which of the statements below?

I. $p \Rightarrow q$
II. $q \Rightarrow p$
III. $(\neg q) \vee p$
IV. $(\neg p) \vee q$

A. I only
B. I and IV only
C. II only
D. II and III only

gatecse-2017-set1 mathematical-logic propositional-logic easy

Answer key☟

3.3.34 Propositional Logic: GATE CSE 2017 | Set 1 | Question: 29

Let p, q and r be propositions and the expression $(p \rightarrow q) \rightarrow r$ be a contradiction. Then, the expression $(r \rightarrow p) \rightarrow q$ is

A. a tautology
B. a contradiction
C. always TRUE when p is FALSE
D. always TRUE when q is TRUE

gatecse-2017-set1 mathematical-logic propositional-logic

Answer key☟

3.3.35 Propositional Logic: GATE CSE 2017 | Set 2 | Question: 11

Let p, q, r denote the statements "*It is raining*", "*It is cold*", and "*It is pleasant*", respectively. Then the statement "*It is not raining and it is pleasant, and it is not pleasant only if it is raining and it is cold*" is represented by

A. $(\neg p \wedge r) \wedge (\neg r \rightarrow (p \wedge q))$
B. $(\neg p \wedge r) \wedge ((p \wedge q) \rightarrow \neg r)$
C. $(\neg p \wedge r) \vee ((p \wedge q) \rightarrow \neg r)$
D. $(\neg p \wedge r) \vee (r \rightarrow (p \wedge q))$

gatecse-2017-set2 mathematical-logic propositional-logic

Answer key

3.3.36 Propositional Logic: GATE CSE 2021 | Set 1 | Question: 7

Let p and q be two propositions. Consider the following two formulae in propositional logic.

- $S_1 : (\neg p \wedge (p \vee q)) \rightarrow q$
- $S_2 : q \rightarrow (\neg p \wedge (p \vee q))$

Which one of the following choices is correct?

A. Both S_1 and S_2 are tautologies.
B. S_1 is a tautology but S_2 is not a tautology
C. S_1 is not a tautology but S_2 is a tautology
D. Neither S_1 nor S_2 is a tautology

gatecse-2021-set1 mathematical-logic propositional-logic one-mark

Answer key

3.3.37 Propositional Logic: GATE CSE 2021 | Set 2 | Question: 15

Choose the correct choice(s) regarding the following proportional logic assertion S:

$$S : ((P \wedge Q) \rightarrow R) \rightarrow ((P \wedge Q) \rightarrow (Q \rightarrow R))$$

A. S is neither a tautology nor a contradiction
B. S is a tautology
C. S is a contradiction
D. The antecedent of S is logically equivalent to the consequent of S

gatecse-2021-set2 multiple-selects mathematical-logic propositional-logic one-mark

Answer key

3.3.38 Propositional Logic: GATE CSE 2024 | Set 2 | Question: 2

Let p and q be the following propositions:

p : Fail grade can be given.
q : Student scores more than 50% marks.

Consider the statement: "*Fail grade cannot be given when student scores more than* 50% *marks.*"

Which one of the following is the CORRECT representation of the above statement in propositional logic?

A. $q \rightarrow \neg p$
B. $q \rightarrow p$
C. $p \rightarrow q$
D. $\neg p \rightarrow q$

gatecse-2024-set2 mathematical-logic propositional-logic one-mark

Answer key

3.3.39 Propositional Logic: GATE DS&AI 2024 | Question: 19

Let x and y be two propositions. Which of the following statements is a tautology /are tautologies?

A. $(\neg x \wedge y) \Rightarrow (y \Rightarrow x)$
B. $(x \wedge \neg y) \Rightarrow (\neg x \Rightarrow y)$
C. $(\neg x \wedge y) \Rightarrow (\neg x \Rightarrow y)$
D. $(x \wedge \neg y) \Rightarrow (y \Rightarrow x)$

gate-ds-ai-2024 mathematical-logic propositional-logic multiple-selects one-mark

Answer key

3.3.40 Propositional Logic: GATE IT 2004 | Question: 31

Let p, q, r and s be four primitive statements. Consider the following arguments:

- $P : [(\neg p \vee q) \wedge (r \rightarrow s) \wedge (p \vee r)] \rightarrow (\neg s \rightarrow q)$
- $Q : [(\neg p \wedge q) \wedge [q \rightarrow (p \rightarrow r)]] \rightarrow \neg r$
- $R : [[(q \wedge r) \rightarrow p] \wedge (\neg q \vee p)] \rightarrow r$
- $S : [p \wedge (p \rightarrow r) \wedge (q \vee \neg r)] \rightarrow q$

Which of the above arguments are valid?

A. P and Q only B. P and R only C. P and S only D. P, Q, R and S

gateit-2004 mathematical-logic normal propositional-logic

Answer key☟

Answer Keys

3.1.1	N/A	3.1.2	N/A	3.1.3	A	3.1.4	D	3.1.5	D
3.1.6	D	3.1.7	B	3.1.8	D	3.1.9	D	3.1.10	X
3.1.11	D	3.1.12	B	3.1.13	B	3.1.14	A	3.1.15	C
3.1.16	D	3.1.17	X	3.1.18	D	3.1.19	D	3.1.20	C
3.1.21	D	3.1.22	B	3.1.23	A	3.1.24	C	3.1.25	C
3.1.26	B;C	3.1.27	B;D	3.1.28	A	3.1.29	A	3.1.30	C
3.1.31	B	3.1.32	B	3.1.33	A	3.1.34	C	3.1.35	D
3.2.1	B	3.2.2	C	3.2.3	A	3.3.1	N/A	3.3.2	N/A
3.3.3	A;B;C	3.3.4	A	3.3.5	B	3.3.6	B	3.3.7	N/A
3.3.8	N/A	3.3.9	B	3.3.10	True	3.3.11	N/A	3.3.12	D
3.3.13	D	3.3.14	C	3.3.15	A	3.3.16	N/A	3.3.17	A
3.3.18	A	3.3.19	A	3.3.20	N/A	3.3.21	B	3.3.22	A
3.3.23	B	3.3.24	D	3.3.25	B	3.3.26	B	3.3.27	B
3.3.28	B	3.3.29	D	3.3.30	C	3.3.31	11	3.3.32	4
3.3.33	D	3.3.34	D	3.3.35	A	3.3.36	B	3.3.37	B;D
3.3.38	A	3.3.39	B;C;D	3.3.40	C				

Syllabus: Sets, Relations, Functions, Partial orders, Lattices, **Monoids,** Groups.

Mark Distribution in Previous GATE

Year	2026 - 1	2026 - 2	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum
1 Mark Count	0	1	1	0	1	1	0	1	0	1	0
2 Marks Count	0	0	1	1	1	1	2	0	2	1	0
Total Marks	0	1	3	2	3	3	4	1	4	3	**0**

Welcome to the "Discrete Mathematics: Set Theory & Algebra" chapter, a cornerstone of computer science that underpins many advanced topics. This section is crucial for the GATE Computer Science exam, typically accounting for **8-12 marks**. Questions often test your foundational understanding of definitions, properties, and the application of theorems, ranging from direct formula recall to multi-step problem-solving involving logical deduction. Mastering these concepts is not just about scoring well; it's about building the analytical and problem-solving skills essential for a career in computing.

Topic-wise Key Concepts

Set Theory

Set theory is the fundamental language for describing collections of objects. It provides the basis for understanding relations, functions, and many other discrete structures.

Definition and Core Idea:

A set is a well-defined collection of distinct objects, called elements. Sets are typically denoted by capital letters, and their elements are enclosed in curly braces. The core idea is to classify and manipulate collections of items.

Important Formulas and Results:

- **Cardinality:** The number of elements in a set A is denoted by $|A|$.
- **Power Set:** The set of all subsets of A, denoted $P(A)$. If $|A| = n$, then $|P(A)| = 2^n$.
- **Principle of Inclusion-Exclusion:**
  - For two sets: $|A \cup B| = |A| + |B| - |A \cap B|$
  - For three sets: $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$

Key Properties and Identities:

- **Commutative Laws:**
  - $A \cup B = B \cup A$
  - $A \cap B = B \cap A$
- **Associative Laws:**
  - $(A \cup B) \cup C = A \cup (B \cup C)$
  - $(A \cap B) \cap C = A \cap (B \cap C)$
- **Distributive Laws:**
  - $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
  - $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- **De Morgan's Laws:**
  - $\overline{A \cup B} = \overline{A} \cap \overline{B}$
  - $\overline{A \cap B} = \overline{A} \cup \overline{B}$
- **Idempotent Laws:**
  - $A \cup A = A$
  - $A \cap A = A$
- **Identity Laws:**
  - $A \cup \emptyset = A$
  - $A \cap U = A$ (where U is the universal set)
- **Complement Laws:**

- $A \cup \overline{A} = U$
- $A \cap \overline{A} = \emptyset$
- $\overline{\emptyset} = U$
- $\overline{U} = \emptyset$
- $\overline{\overline{A}} = A$

Common Pitfalls & Tricky Points:

- Confusing elements with sets containing those elements (e.g., $a \in \{a\}$ but $a \neq \{a\}$).
- Misinterpreting the empty set $\emptyset$. Remember $\emptyset$ is a subset of every set, and $P(\emptyset) = \{\emptyset\}$.
- Errors in applying De Morgan's Laws or Inclusion-Exclusion for complex scenarios.

Problem-Solving Techniques:

- Use Venn diagrams for visualizing set operations and proving identities for up to three sets.
- For formal proofs, use element-wise arguments (e.g., to prove $A \subseteq B$, show that if $x \in A$, then $x \in B$).
- Apply the Principle of Inclusion-Exclusion systematically for counting problems involving overlapping sets.

Relations

A relation describes how elements from one set relate to elements of another set (or the same set). It's a fundamental concept for modeling connections and dependencies.

Definition and Core Idea:

A relation R from set A to set B is a subset of the Cartesian product $A \times B$. If $(a, b) \in R$, we say a is related to b, often written aRb. The core idea is to define specific associations between elements.

Important Formulas and Results:

- **Number of relations:** If $|A| = m$ and $|B| = n$, the number of relations from A to B is 2^{mn}.
- **Types of Relations on a Set A:**
  1. **Reflexive:** For all $a \in A$, $(a, a) \in R$.
  2. **Irreflexive:** For all $a \in A$, $(a, a) \notin R$.
  3. **Symmetric:** For all $a, b \in A$, if $(a, b) \in R$, then $(b, a) \in R$.
  4. **Antisymmetric:** For all $a, b \in A$, if $(a, b) \in R$ and $(b, a) \in R$, then $a = b$.
  5. **Asymmetric:** For all $a, b \in A$, if $(a, b) \in R$, then $(b, a) \notin R$. (Implies irreflexive and antisymmetric).
  6. **Transitive:** For all $a, b, c \in A$, if $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$.
- **Equivalence Relation:** A relation that is Reflexive, Symmetric, and Transitive.
- **Equivalence Class:** If R is an equivalence relation on A, the equivalence class of an element $a \in A$ is $[a]_R = \{x \in A \mid (x, a) \in R\}$. Equivalence classes partition the set A.
- **Partial Order Relation:** A relation that is Reflexive, Antisymmetric, and Transitive. (See Partial Order section for more details).

Key Properties and Identities:

- The inverse of a relation R is $R^{-1} = \{(b, a) \mid (a, b) \in R\}$.
- Composition of relations: If $R_1 : A \to B$ and $R_2 : B \to C$, then $R_2 \circ R_1 = \{(a, c) \mid \exists b \in B, (a, b) \in R_1 \text{ and } (b, c) \in R_2\}$.

Common Pitfalls & Tricky Points:

- Confusing symmetric with antisymmetric. A relation can be neither, both (only if it contains only pairs (a, a)), or one but not the other.
- Verifying transitivity requires checking all possible chains (a, b) and (b, c).
- Forgetting to check reflexivity for equivalence or partial order relations.

Problem-Solving Techniques:

- Represent relations using directed graphs (digraphs) or adjacency matrices, especially for small sets.
- To prove a relation is of a certain type, systematically check each property by definition.
- For equivalence relations, identify the equivalence classes to understand the partition of the set.

Functions

Functions are special types of relations where each input maps to exactly one output. They are central to all areas of mathematics and computer science.

Definition and Core Idea:

A function f from set A (domain) to set B (codomain), denoted $f : A \to B$, assigns to each element $x \in A$ exactly one element $y \in B$. The core idea is a deterministic mapping from inputs to outputs.

Important Formulas and Results:

- **Range:** The set of all actual output values, $\text{range}(f) = \{f(x) \mid x \in A\}$. Note that $\text{range}(f) \subseteq B$.
- **Types of Functions:**
  1. **Injective (One-to-one):** If $f(x_1) = f(x_2) \implies x_1 = x_2$ for all $x_1, x_2 \in A$. Each element in the codomain is mapped to by at most one element in the domain.
  2. **Surjective (Onto):** For every $y \in B$, there exists at least one $x \in A$ such that $f(x) = y$. The range of f is equal to its codomain B.
  3. **Bijective (One-to-one and Onto):** A function that is both injective and surjective.
- **Inverse Function:** A function $f : A \to B$ has an inverse function $f^{-1} : B \to A$ if and only if f is bijective. Then $f^{-1}(y) = x \iff f(x) = y$.
- **Composition of Functions:** If $f : A \to B$ and $g : B \to C$, then $(g \circ f) : A \to C$ is defined as $(g \circ f)(x) = g(f(x))$.
- **Number of Functions:** If $|A| = m$ and $|B| = n$:
  - Total functions from A to B: n^m
  - Injective functions from A to B: $P(n, m) = \frac{n!}{(n-m)!}$ if $m \leq n$, else 0.
  - Bijective functions from A to B: $n!$ if $m = n$, else 0.
  - Surjective functions from A to B: $n! \cdot S(m, n)$, where $S(m, n)$ is the Stirling number of the second kind. Alternatively, using Inclusion-Exclusion: $\sum_{k=0}^{n} (-1)^k \binom{n}{k} (n-k)^m$.

Key Properties and Identities:

- Composition of functions is associative: $(h \circ g) \circ f = h \circ (g \circ f)$.
- If f is bijective, then $f^{-1} \circ f = I_A$ and $f \circ f^{-1} = I_B$, where I_A and I_B are identity functions on sets A and B respectively.

Common Pitfalls & Tricky Points:

- Confusing the codomain with the range.
- Incorrectly assuming a function has an inverse when it's not bijective.
- Proving surjectivity requires showing that *every* element in the codomain has a preimage, not just some.

Problem-Solving Techniques:

- To prove injectivity, assume $f(x_1) = f(x_2)$ and algebraically show $x_1 = x_2$.
- To prove surjectivity, take an arbitrary y from the codomain and show there exists an x in the domain such that $f(x) = y$.
- When counting functions, carefully identify the constraints (injective, surjective, bijective).

Binary Operation

A binary operation combines two elements from a set to produce a third element within the same set. It's the foundation for algebraic structures like groups and rings.

Definition and Core Idea:

A binary operation $*$ on a set S is a function $*: S \times S \to S$. For any $a, b \in S$, $a * b$ is a unique element in S. The core idea is to define an internal way to combine elements.

Important Formulas and Results:

- **Properties of a Binary Operation:**
  1. **Closure:** For all $a, b \in S$, $a * b \in S$. (This is inherent in the definition).
  2. **Associativity:** For all $a, b, c \in S$, $(a * b) * c = a * (b * c)$.
  3. **Commutativity:** For all $a, b \in S$, $a * b = b * a$.
  4. **Identity Element:** An element $e \in S$ such that for all $a \in S$, $a * e = e * a = a$. If it exists, it is unique.
  5. **Inverse Element:** For an element $a \in S$, an element $a^{-1} \in S$ is its inverse if $a * a^{-1} = a^{-1} * a = e$, where e is the identity element. If an identity exists, inverses are unique for each element.

Key Properties and Identities:

- If an identity element exists, it is unique.
- If an identity element exists and the operation is associative, then inverses are unique.

Common Pitfalls & Tricky Points:

- Forgetting to check closure: the result of the operation must always be in the original set.
- Misidentifying the identity element or assuming it always exists.
- Assuming commutativity or associativity when not explicitly stated or proven.

Problem-Solving Techniques:

- To check properties, test with arbitrary elements or counterexamples.
- For finite sets, construct a Cayley table (operation table) to check properties visually.

Group Theory

Group theory studies algebraic structures called groups, which are sets equipped with a binary operation satisfying specific axioms. It's fundamental in cryptography, coding theory, and physics.

Definition and Core Idea:

A group $(G, *)$ is a set G together with a binary operation $*$ that satisfies four axioms:

1. **Closure:** For all $a, b \in G$, $a * b \in G$.
2. **Associativity:** For all $a, b, c \in G$, $(a * b) * c = a * (b * c)$.
3. **Identity Element:** There exists an element $e \in G$ such that for all $a \in G$, $a * e = e * a = a$.
4. **Inverse Element:** For each $a \in G$, there exists an element $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$.

The core idea is to formalize symmetry and transformations.

Important Formulas and Results:

- **Related Structures:**
  - **Semigroup:** A set with an associative binary operation.
  - **Monoid:** A semigroup with an identity element.
  - **Abelian Group (Commutative Group):** A group where the operation is commutative ($a * b = b * a$).
- **Subgroup:** A non-empty subset H of a group G is a subgroup if H is itself a group under the same operation.
  - **Subgroup Test:** H is a subgroup of G if for all $a, b \in H$, $a * b^{-1} \in H$.
- **Cyclic Group:** A group G is cyclic if there exists an element $g \in G$ such that every element of G can be written as a power of g (i.e., $G = \{g^n \mid n \in \mathbb{Z}\}$). g is called a generator.
- **Order of an element:** The smallest positive integer n such that $a^n = e$. If no such n exists, the element has infinite

order.

- **Lagrange's Theorem:** If G is a finite group and H is a subgroup of G, then the order of H divides the order of G (i.e., $|H|$ divides $|G|$).

Key Properties and Identities:

- The identity element is unique.
- Every element has a unique inverse.
- $(a^{-1})^{-1} = a$
- $(a * b)^{-1} = b^{-1} * a^{-1}$

Common Pitfalls & Tricky Points:

- Forgetting to check all four group axioms, especially the existence of inverses for *every* element.
- Confusing the order of an element with the order of the group.
- Misapplying Lagrange's Theorem (it states a necessary condition for a subgroup, not a sufficient one).

Problem-Solving Techniques:

- To prove a set with an operation is a group, systematically verify all four axioms.
- To find subgroups, look for subsets that are closed under the operation and contain the identity and inverses.
- Use Lagrange's Theorem to quickly rule out potential subgroup orders.

Identity Function

The identity function is a fundamental concept in functions and algebraic structures, acting as a neutral element for function composition.

Definition and Core Idea:

For any set A, the identity function on A, denoted I_A or id_A, is a function $I_A : A \to A$ such that for every element $x \in A$, $I_A(x) = x$. The core idea is a mapping that leaves elements unchanged.

Important Formulas and Results:

- If $f : A \to B$ is any function, then $f \circ I_A = f$ and $I_B \circ f = f$.
- If $f : A \to B$ is a bijective function, then its inverse $f^{-1} : B \to A$ satisfies $f^{-1} \circ f = I_A$ and $f \circ f^{-1} = I_B$.

Key Properties and Identities:

- The identity function is always bijective (one-to-one and onto).
- It serves as the identity element for the operation of function composition.

Common Pitfalls & Tricky Points:

- Confusing the identity function with a constant function (e.g., $f(x) = c$).
- Forgetting that the identity function is specific to a set A.

Problem-Solving Techniques:

- Use the properties of the identity function when simplifying compositions or proving inverse relationships.

Lattice

Lattices are special types of partially ordered sets where every pair of elements has a unique "least upper bound" and "greatest lower bound." They are crucial in areas like logic, set theory, and computer science for modeling hierarchical structures.

Definition and Core Idea:

A poset $(L, \preceq)$ is a lattice if for every pair of elements $a, b \in L$, both their least upper bound (LUB or join, denoted $a \vee b$) and greatest lower bound (GLB or meet, denoted $a \wedge b$) exist and are unique. The core idea is a structure where elements always have well-defined "closest" common superiors and inferiors.

Important Formulas and Results:

- **Join (Supremum):** $a \vee b = \sup\{a, b\}$.
- **Meet (Infimum):** $a \wedge b = \inf\{a, b\}$.
- **Types of Lattices:**
  - **Distributive Lattice:** A lattice where the distributive laws hold:
    - $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$
    - $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$
  - **Complemented Lattice:** A bounded lattice (has a greatest element 1 and a least element 0) where every element a has a complement a' such that $a \wedge a' = 0$ and $a \vee a' = 1$.
  - **Boolean Algebra:** A distributive and complemented lattice.

Key Properties and Identities:

- **Idempotent Laws:** $a \wedge a = a$, $a \vee a = a$
- **Commutative Laws:** $a \wedge b = b \wedge a$, $a \vee b = b \vee a$
- **Associative Laws:** $(a \wedge b) \wedge c = a \wedge (b \wedge c)$, $(a \vee b) \vee c = a \vee (b \vee c)$
- **Absorption Laws:** $a \wedge (a \vee b) = a$, $a \vee (a \wedge b) = a$
- **Duality Principle:** If a statement is true for a lattice, its dual (exchanging $\wedge$ with $\vee$, $\preceq$ with $\succeq$, 0 with 1) is also true.

Common Pitfalls & Tricky Points:

- Not every poset is a lattice. A poset fails to be a lattice if any pair of elements lacks a unique LUB or GLB.
- Distinguishing between maximal/minimal elements and greatest/least elements in a poset (a lattice always has a greatest and least element if it's bounded).

Problem-Solving Techniques:

- Draw Hasse diagrams to visually identify LUBs and GLBs for all pairs.
- To check for distributivity, look for sublattices isomorphic to N_5 (pentagon lattice) or M_3 (diamond lattice) in the Hasse diagram (they are non-distributive).

Mathematical Induction

Mathematical Induction is a powerful proof technique used to establish the truth of a statement for all natural numbers (or a subset thereof). It's essential for proving properties of algorithms, data structures, and number theory results.

Definition and Core Idea:

The Principle of Mathematical Induction states that to prove a statement $P(n)$ is true for all integers $n \geq n_0$:

1. **Base Case:** Show that $P(n_0)$ is true.
2. **Inductive Hypothesis:** Assume that $P(k)$ is true for some arbitrary integer $k \geq n_0$.
3. **Inductive Step:** Show that $P(k+1)$ is true, using the inductive hypothesis.

The core idea is like a chain reaction: if the first domino falls, and falling dominoes always knock over the next one, then all dominoes will fall.

Important Formulas and Results:

- **Principle of Strong Induction:**
  1. **Base Case:** Show that $P(n_0)$ is true.
  2. **Inductive Hypothesis:** Assume that $P(j)$ is true for all integers j such that $n_0 \leq j \leq k$, for some arbitrary integer $k \geq n_0$.
  3. **Inductive Step:** Show that $P(k+1)$ is true, using the inductive hypothesis.

Key Properties and Identities:

- Both standard and strong induction are equivalent in power; strong induction is often more convenient for proofs where $P(k+1)$ depends on more than just $P(k)$.

Common Pitfalls & Tricky Points:

- Incorrect base case: The statement must hold for the starting value.
- Assuming what needs to be proven in the inductive step. The inductive hypothesis must be explicitly used.
- Not clearly stating the inductive hypothesis or the inductive step.
- Algebraic errors during the inductive step.

Problem-Solving Techniques:

- Clearly label each step: Base Case, Inductive Hypothesis, Inductive Step.
- In the inductive step, try to manipulate the expression for $P(k+1)$ to isolate an instance of $P(k)$ that can be substituted using the hypothesis.
- For inequalities, sometimes it's easier to prove $P(k+1) - P(k) \geq 0$ or similar.

Number Theory

Number theory is the study of integers and their properties, including divisibility, prime numbers, and modular arithmetic. It has direct applications in cryptography, error correction codes, and algorithm analysis.

Definition and Core Idea:

Number theory explores the relationships and properties of integers. The core idea is to understand the structure and behavior of numbers under various operations.

Important Formulas and Results:

- **Divisibility:** $a|b$ means a divides b (i.e., $b = ka$ for some integer k).
- **Division Algorithm:** For integers a (dividend) and d (divisor) with $d > 0$, there exist unique integers q (quotient) and r (remainder) such that $a = dq + r$, where $0 \leq r < d$.
- **Greatest Common Divisor (GCD):** $\gcd(a, b)$ is the largest integer that divides both a and b.
  - **Euclidean Algorithm:** An efficient method to compute GCD. $\gcd(a, b) = \gcd(b, a \pmod b)$.
  - **Bézout's Identity:** For any integers a, b, there exist integers x, y such that $ax + by = \gcd(a, b)$.
- **Least Common Multiple (LCM):** $\operatorname{lcm}(a, b)$ is the smallest positive integer divisible by both a and b.
  - Relationship: $\gcd(a, b) \cdot \operatorname{lcm}(a, b) = |ab|$.
- **Prime Numbers:** A positive integer greater than 1 with exactly two positive divisors: 1 and itself.
  - **Fundamental Theorem of Arithmetic:** Every integer greater than 1 can be uniquely represented as a product of prime numbers (up to the order of factors).
- **Modular Arithmetic:** $a \equiv b \pmod m$ means $m|(a-b)$. This is an equivalence relation.
  - Properties:
    - If $a \equiv b \pmod m$ and $c \equiv d \pmod m$, then $a + c \equiv b + d \pmod m$ and $ac \equiv bd \pmod m$.
    - $(a \pmod m) + b \pmod m) \pmod m = (a + b) \pmod m$
    - $(a \pmod m) \cdot b \pmod m) \pmod m = (a \cdot b) \pmod m$
- **Fermat's Little Theorem:** If p is a prime number and a is an integer not divisible by p, then $a^{p-1} \equiv 1 \pmod p$. Also, $a^p \equiv a \pmod p$ for any integer a.
- **Euler's Totient Function $\phi(n)$:** Counts the number of positive integers up to n that are relatively prime to n.
  - If $n = p_1^{k_1} p_2^{k_2} \dots p_r^{k_r}$ is the prime factorization of n, then $\phi(n) = n\left(1 - \frac{1}{p_1}\right)\left(1 - \frac{1}{p_2}\right)\dots\left(1 - \frac{1}{p_r}\right)$.
  - If p is prime, $\phi(p) = p - 1$.
- **Euler's Theorem:** If n is a positive integer and a is an integer relatively prime to n (i.e., $\gcd(a, n) = 1$), then $a^{\phi(n)} \equiv 1 \pmod n$.

Key Properties and Identities:

- Congruence modulo m is an equivalence relation.

- The set of integers modulo n, $\mathbb{Z}_n = \{0, 1, \dots, n-1\}$, forms a ring under addition and multiplication modulo n. If n is prime, $\mathbb{Z}_n$ is a field.

Common Pitfalls & Tricky Points:

- Mistakes in applying modular arithmetic, especially with negative numbers or division. Remember division is not always well-defined in modular arithmetic (requires modular inverse).
- Confusing prime numbers with relatively prime numbers.
- Incorrectly calculating $\phi(n)$ for composite numbers.

Problem-Solving Techniques:

- Use the Euclidean algorithm for GCD and extended Euclidean algorithm for modular inverses.
- Simplify large powers in modular arithmetic using Fermat's Little Theorem or Euler's Theorem.
- For divisibility problems, use properties of congruences.

Onto (Surjective Functions)

Onto functions, also known as surjective functions, ensure that every element in the codomain is "hit" by at least one element from the domain. This property is vital for understanding mappings that cover their entire target set.

Definition and Core Idea:

A function $f: A \to B$ is said to be onto (or surjective) if for every element $y \in B$ (the codomain), there exists at least one element $x \in A$ (the domain) such that $f(x) = y$. The core idea is that the function's range completely fills its codomain.

Important Formulas and Results:

- A function $f: A \to B$ is surjective if and only if $\text{range}(f) = B$.
- If A and B are finite sets, and $f: A \to B$ is surjective, then $|A| \geq |B|$.
- The number of surjective functions from a set of size m to a set of size n is given by $\sum_{k=0}^{n} (-1)^k \binom{n}{k} (n-k)^m$.

Key Properties and Identities:

- If f and g are surjective, then $g \circ f$ is surjective.
- If $g \circ f$ is surjective, then g must be surjective (but f need not be).

Common Pitfalls & Tricky Points:

- Confusing surjectivity with injectivity. An onto function doesn't require unique preimages.
- Failing to demonstrate a preimage for an arbitrary element in the codomain, instead just showing some elements have preimages.

Problem-Solving Techniques:

- To prove a function $f: A \to B$ is surjective, pick an arbitrary $y \in B$, then show how to construct an $x \in A$ such that $f(x) = y$. This often involves solving $y = f(x)$ for x.
- For finite sets, compare the cardinalities of the domain and codomain. If $|A| < |B|$, no surjective function from A to B can exist.

Partial Order

Partial order relations define a sense of relative order among elements in a set, but not necessarily for every pair. They are crucial for modeling hierarchies, dependencies, and precedence in computer science (e.g., task scheduling, class inheritance).

Definition and Core Idea:

A relation R on a set A is a partial order relation if it is reflexive, antisymmetric, and transitive. A set A with a partial order relation $\preceq$ is called a partially ordered set (poset), denoted $(A, \preceq)$. The core idea is to establish an ordering where some elements might be incomparable.

Important Formulas and Results:

- **Properties of Partial Order:**
  1. **Reflexive:** For all $a \in A$, $a \preceq a$.
  2. **Antisymmetric:** For all $a, b \in A$, if $a \preceq b$ and $b \preceq a$, then $a = b$.
  3. **Transitive:** For all $a, b, c \in A$, if $a \preceq b$ and $b \preceq c$, then $a \preceq c$.
- **Comparability:** Two elements $a, b \in A$ are comparable if $a \preceq b$ or $b \preceq a$. Otherwise, they are incomparable.
- **Total Order (Linear Order):** A partial order where every pair of elements is comparable.
- **Hasse Diagram:** A graphical representation of a finite poset, where:
  - Elements are represented by nodes.
  - An edge goes upwards from a to b if $a \preceq b$ and there is no c such that $a \preceq c \preceq b$ (i.e., b covers a).
  - Reflexive loops and transitive edges are omitted.
- **Elements in a Poset:**
  - **Maximal Element:** An element a such that there is no $b \in A$ with $a \preceq b$ and $a \neq b$.
  - **Minimal Element:** An element a such that there is no $b \in A$ with $b \preceq a$ and $a \neq b$.
  - **Greatest Element (Maximum):** An element g such that for all $x \in A$, $x \preceq g$. (If it exists, it is unique).
  - **Least Element (Minimum):** An element l such that for all $x \in A$, $l \preceq x$. (If it exists, it is unique).
  - **Upper Bound:** For a subset $S \subseteq A$, an element $u \in A$ is an upper bound of S if $s \preceq u$ for all $s \in S$.
  - **Lower Bound:** For a subset $S \subseteq A$, an element $l \in A$ is a lower bound of S if $l \preceq s$ for all $s \in S$.
  - **Least Upper Bound (LUB / Supremum):** The least element among all upper bounds of S, denoted $\sup(S)$.
  - **Greatest Lower Bound (GLB / Infimum):** The greatest element among all lower bounds of S, denoted $\inf(S)$.

Key Properties and Identities:

- A greatest element is always a maximal element, but a maximal element is not necessarily the greatest. Similarly for least and minimal elements.
- A finite poset always has at least one maximal and one minimal element.

Common Pitfalls & Tricky Points:

- Confusing maximal/minimal elements with greatest/least elements. A poset can have multiple maximal elements but at most one greatest element.
- Errors in drawing Hasse diagrams, especially omitting necessary edges or including redundant ones.
- Incorrectly identifying LUBs or GLBs for subsets.

Problem-Solving Techniques:

- To verify a partial order, systematically check reflexivity, antisymmetry, and transitivity.
- Draw Hasse diagrams to visualize the order and identify special elements (maximal, minimal, greatest, least, LUB, GLB).
- For LUB/GLB, first find all upper/lower bounds, then identify the least/greatest among them.

Polynomials

Polynomials are fundamental algebraic expressions used extensively in computer science for modeling, interpolation, error correction, and algorithm analysis (e.g., complexity). In discrete mathematics, they often appear in contexts like finite fields and generating functions.

Definition and Core Idea:

A polynomial in a variable x is an expression of the form $P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$, where a_i are coefficients (often real or complex numbers, but can be from finite fields in discrete math) and n is a non-negative integer called the degree of the polynomial (if $a_n \neq 0$). The core idea is to represent relationships using sums of powers of a variable.

Important Formulas and Results:

- **Degree:** The highest power of x with a non-zero coefficient.
- **Root (or Zero):** A value c such that $P(c) = 0$.
- **Polynomial Division Algorithm:** For polynomials $P(x)$ (dividend) and $D(x)$ (divisor) with $D(x) \neq 0$, there exist unique polynomials $Q(x)$ (quotient) and $R(x)$ (remainder) such that $P(x) = D(x)Q(x) + R(x)$, where $\deg(R(x)) < \deg(D(x))$ or $R(x) = 0$.
- **Remainder Theorem:** If a polynomial $P(x)$ is divided by $(x - c)$, the remainder is $P(c)$.
- **Factor Theorem:** $(x - c)$ is a factor of $P(x)$ if and only if $P(c) = 0$ (i.e., c is a root of $P(x)$).
- **Fundamental Theorem of Algebra:** Every non-constant polynomial with complex coefficients has at least one complex root. Consequently, a polynomial of degree n has exactly n complex roots (counting multiplicity).
- **Vieta's Formulas:** Relate the coefficients of a polynomial to the sums and products of its roots. For a quadratic $ax^2 + bx + c = 0$ with roots r_1, r_2: $r_1 + r_2 = -b/a$, $r_1 r_2 = c/a$.

Key Properties and Identities:

- A polynomial of degree n has at most n distinct roots.
- Polynomials form a ring under addition and multiplication.

Common Pitfalls & Tricky Points:

- Algebraic errors in polynomial division or evaluating polynomials.
- Forgetting to count root multiplicities when applying the Fundamental Theorem of Algebra.
- Not considering the domain of coefficients (e.g., real vs. complex vs. finite field).

Problem-Solving Techniques:

- Use synthetic division for quick division by $(x - c)$.
- Apply the Remainder and Factor Theorems to find roots or factors efficiently.
- For problems involving roots, Vieta's formulas can often provide shortcuts.

Quick Formula Reference

- **Set Theory:**
  - $|P(A)| = 2^{|A|}$
  - $|A \cup B| = |A| + |B| - |A \cap B|$
  - De Morgan's Laws: $\overline{A \cup B} = \overline{A} \cap \overline{B}$, $\overline{A \cap B} = \overline{A} \cup \overline{B}$
- **Relations:**
  - Number of relations from A to B: $2^{|A||B|}$
  - Equivalence Relation: Reflexive, Symmetric, Transitive.
  - Partial Order: Reflexive, Antisymmetric, Transitive.
- **Functions (from $|A| = m$ to $|B| = n$):**
  - Total functions: n^m
  - Injective functions: $P(n, m) = \frac{n!}{(n-m)!}$ (if $m \leq n$)
  - Bijective functions: $n!$ (if $m = n$)
  - Surjective functions: $\sum_{k=0}^{n} (-1)^k \binom{n}{k} (n-k)^m$
  - Identity function $I_A(x) = x$
- **Group Theory:**
  - Group Axioms: Closure, Associativity, Identity, Inverse.
  - Lagrange's Theorem: $|H|$ divides $|G|$ for subgroup H of group G.
- **Lattice:**
  - Poset where every pair has unique LUB (join $\vee$) and GLB (meet $\wedge$).
  - Distributive Lattice: $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$, etc.
- **Mathematical Induction:**
  - Base Case $P(n_0)$, Inductive Hypothesis $P(k)$, Inductive Step $P(k+1)$.
- **Number Theory:**
  - Division Algorithm: $a = dq + r, 0 \leq r < d$
  - $\gcd(a, b) \cdot \operatorname{lcm}(a, b) = |ab|$
  - Fermat's Little Theorem: $a^{p-1} \equiv 1 \pmod{p}$ (for prime p, $p \nmid a$)

- Euler's Totient Function: $\phi(n) = n \prod_{p|n, p \text{ prime}} \left(1 - \frac{1}{p}\right)$
- Euler's Theorem: $a^{\phi(n)} \equiv 1 \pmod{n}$ (for $\gcd(a, n) = 1$)

- **Polynomials:**
  - Remainder Theorem: $P(x) \pmod{(x - c)} = P(c)$
  - Factor Theorem: $(x - c)$ is a factor $\iff P(c) = 0$

Important Tips for GATE

1. **Master Definitions and Properties:** Many GATE questions directly test your understanding of definitions (e.g., what makes a relation symmetric?) and properties (e.g., De Morgan's laws). Memorize them thoroughly.
2. **Practice with Hasse Diagrams:** For partial orders and lattices, drawing Hasse diagrams is crucial. Practice identifying maximal/minimal elements, greatest/least elements, and LUB/GLB from diagrams.
3. **Systematic Verification:** When asked to check if a structure is a group, a relation is an equivalence relation, or a function is bijective, systematically go through each axiom/property. Don't skip steps or assume properties.
4. **Number Theory Shortcuts:** Familiarize yourself with the Euclidean algorithm for GCD, modular arithmetic properties, and theorems like Fermat's Little Theorem and Euler's Theorem. These are vital for efficiently solving problems involving large numbers or powers.
5. **Inclusion-Exclusion Principle:** This principle is frequently tested for counting problems in set theory and functions (especially surjective functions). Practice applying it for 2, 3, and general cases.
6. **Mathematical Induction Structure:** Always clearly state your base case, inductive hypothesis, and inductive step. Ensure your inductive step explicitly uses the inductive hypothesis to avoid circular reasoning.
7. **Cardinality vs. Countability:** Understand the difference between finite, countably infinite, and uncountable sets. Be familiar with standard examples (e.g., $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ are countable; $\mathbb{R}, P(\mathbb{N})$ are uncountable) and Cantor's diagonalization argument.
8. **Watch for "If and Only If":** Pay close attention to conditions that are "if and only if" (iff) as they imply equivalence and are often key to proving properties or identifying specific structures.

4.1 Binary Operation (8)

4.1.1 Binary Operation: GATE CSE 1989 | Question: 1-v

The number of possible commutative binary operations that can be defined on a set of n elements (for a given n) is ____________.

gate1989 descriptive set-theory&algebra binary-operation

Answer key

4.1.2 Binary Operation: GATE CSE 1994 | Question: 2.2

On the set N of non-negative integers, the binary operation _______ is associative and non-commutative.

gate1994 set-theory&algebra normal group-theory binary-operation fill-in-the-blanks

Answer key

4.1.3 Binary Operation: GATE CSE 2003 | Question: 38

Consider the set $\{a, b, c\}$ with binary operators $+$ and $*$ defined as follows:

$+$	**a**	**b**	**c**
a	b	a	c
b	a	b	c
c	a	c	b

$*$	**a**	**b**	**c**
a	a	b	c
b	b	c	a
c	c	c	b

For example, $a + c = c, c + a = a, c * b = c$ and $b * c = a$.

Given the following set of equations:

- $(a * x) + (a * y) = c$
- $(b * x) + (c * y) = c$

The number of solution(s) (i.e., pair(s) (x, y) that satisfy the equations) is

A. 0 B. 1 C. 2 D. 3

gatecse-2003 set-theory&algebra normal binary-operation

Answer key

4.1.4 Binary Operation: GATE CSE 2006 | Question: 28

A logical binary relation $\odot$, is defined as follows:

A	**B**	**A ⊙ B**
True	True	True
True	False	True
False	True	False
False	False	True

Let $\sim$ be the unary negation (NOT) operator, with higher precedence then $\odot$.
Which one of the following is equivalent to $A \wedge B$?

A. $(\sim A \odot B)$
B. $\sim (A \odot \sim B)$
C. $\sim (\sim A \odot \sim B)$
D. $\sim (\sim A \odot B)$

gatecse-2006 set-theory&algebra binary-operation propositional-logic mathematical-logic

Answer key

4.1.5 Binary Operation: GATE CSE 2013 | Question: 1

A binary operation $\oplus$ on a set of integers is defined as $x \oplus y = x^2 + y^2$. Which one of the following statements is **TRUE** about $\oplus$?

A. Commutative but not associative
B. Both commutative and associative
C. Associative but not commutative
D. Neither commutative nor associative

gatecse-2013 set-theory&algebra easy binary-operation

Answer key

4.1.6 Binary Operation: GATE CSE 2015 | Set 1 | Question: 28

The binary operator $\neq$ is defined by the following truth table.

p	**q**	**p ≠ q**
0	0	0
0	1	1
1	0	1
1	1	0

Which one of the following is true about the binary operator $\neq$?

A. Both commutative and associative
B. Commutative but not associative
C. Not commutative but associative
D. Neither commutative nor associative

gatecse-2015-set1 set-theory&algebra easy binary-operation boolean-algebra

Answer key

4.1.7 Binary Operation: GATE CSE 2015 | Set 3 | Question: 2

Let $\#$ be the binary operator defined as
$X\#Y = X' + Y'$ where X and Y are Boolean variables.
Consider the following two statements.

- (S_1) $(P\#Q)\#R = P\#(Q\#R)$

- $(S_2)\ Q\#R = (R\#Q)$

Which are the following is/are true for the Boolean variables P, Q and R?

A. Only S_1 is true
B. Only S_2 is true
C. Both S_1 and S_2 are true
D. Neither S_1 nor S_2 are true

gatecse-2015-set3 set-theory&algebra binary-operation normal

Answer key

4.1.8 Binary Operation: GATE IT 2006 | Question: 2

For the set N of natural numbers and a binary operation $f : N \times N \rightarrow N$, an element $z \in N$ is called an identity for f, if $f(a, z) = a = f(z, a)$, for all $a \in N$. Which of the following binary operations have an identity?

I. $f(x, y) = x + y - 3$
II. $f(x, y) = \max(x, y)$
III. $f(x, y) = x^y$

A. I and II only
B. II and III only
C. I and III only
D. None of these

gateit-2006 set-theory&algebra easy binary-operation

Answer key

4.2 Countable Uncountable Set (2)

4.2.1 Countable Uncountable Set: GATE CSE 1994 | Question: 3.9

Every subset of a countable set is countable.

State whether the above statement is true or false with reason.

gate1994 set-theory&algebra normal set-theory countable-uncountable-set true-false

Answer key

4.2.2 Countable Uncountable Set: GATE CSE 2018 | Question: 27

Let N be the set of natural numbers. Consider the following sets,

- P : Set of Rational numbers (positive and negative)
- Q : Set of functions from $\{0, 1\}$ to N
- R : Set of functions from N to $\{0, 1\}$
- S : Set of finite subsets of N

Which of the above sets are countable?

A. Q and S only
B. P and S only
C. P and R only
D. P, Q and S only

gatecse-2018 set-theory&algebra countable-uncountable-set normal two-marks

Answer key

4.3 Functions (30)

4.3.1 Functions: GATE CSE 1987 | Question: 9b

How many one-to-one functions are there from a set A with n elements onto itself?

gate1987 set-theory&algebra functions descriptive

Answer key

4.3.2 Functions: GATE CSE 1988 | Question: 13ii

If the set S has a finite number of elements, prove that if f maps S onto S, then f is one-to-one.

gate1988 descriptive set-theory&algebra functions

Answer key

4.3.3 Functions: GATE CSE 1989 | Question: 13c

Find the number of single valued functions from set A to another set B, given that the cardinalities of the sets A and B are m and n respectively.

gate1989 descriptive functions set-theory&algebra

Answer key

4.3.4 Functions: GATE CSE 1993 | Question: 8.6

Let A and B be sets with cardinalities m and n respectively. The number of one-one mappings from A to B, when $m < n$, is

A. m^n
B. ${}^nP_m$
C. ${}^mC_n$
D. ${}^nC_m$
E. ${}^mP_n$

gate1993 set-theory&algebra functions easy

Answer key

4.3.5 Functions: GATE CSE 1996 | Question: 1.3

Suppose X and Y are sets and $|X|$ and $|Y|$ are their respective cardinality. It is given that there are exactly 97 functions from X to Y. From this one can conclude that

A. $|X| = 1, |Y| = 97$
B. $|X| = 97, |Y| = 1$
C. $|X| = 97, |Y| = 97$
D. None of the above

gate1996 set-theory&algebra functions normal

Answer key

4.3.6 Functions: GATE CSE 1996 | Question: 2.1

Let R denote the set of real numbers. Let $f : R \times R \rightarrow R \times R$ be a bijective function defined by $f(x, y) = (x + y, x - y)$. The inverse function of f is given by

A. $f^{-1}(x,y) = \left(\frac{1}{x+y}, \frac{1}{x-y}\right)$
B. $f^{-1}(x,y) = (x - y, x + y)$
C. $f^{-1}(x,y) = \left(\frac{x+y}{2}, \frac{x-y}{2}\right)$
$f^{-1}(x,y) = [2(x - y), 2(x + y)]$

gate1996 set-theory&algebra functions normal

Answer key

4.3.7 Functions: GATE CSE 1997 | Question: 13

Let F be the set of one-to-one functions from the set $\{1, 2, \dots, n\}$ to the set $\{1, 2, \dots, m\}$ where $m \geq n \geq 1$.

a. How many functions are members of F?
b. How many functions f in F satisfy the property $f(i) = 1$ for some $i, 1 \leq i \leq n$?
c. How many functions f in F satisfy the property $f(i) < f(j)$ for all i, j $1 \leq i \leq j \leq n$?

gate1997 set-theory&algebra functions normal descriptive difficult

Answer key

4.3.8 Functions: GATE CSE 1998 | Question: 1.8

The number of functions from an m element set to an n element set is

A. $m+n$ B. m^n C. n^m D. $m*n$

gate1998 set-theory&algebra combinatory functions easy

Answer key

4.3.9 Functions: GATE CSE 2001 | Question: 2.3

Let $f : A \to B$ a function, and let E and F be subsets of A. Consider the following statements about images.

- $S_1 : f(E \cup F) = f(E) \cup f(F)$
- $S_2 : f(E \cap F) = f(E) \cap f(F)$

Which of the following is true about S1 and S2?

A. Only S_1 is correct
B. Only S_2 is correct
C. Both S_1 and S_2 are correct
D. None of S_1 and S_2 is correct

gatecse-2001 set-theory&algebra functions normal

Answer key

4.3.10 Functions: GATE CSE 2001 | Question: 4

Consider the function $h : N \times N \to N$ so that $h(a,b) = (2a+1)2^b - 1$, where $N = \{0,1,2,3,\dots\}$ is the set of natural numbers.

a. Prove that the function h is an injection (one-one).
b. Prove that it is also a Surjection (onto)

gatecse-2001 functions set-theory&algebra normal descriptive

Answer key

4.3.11 Functions: GATE CSE 2003 | Question: 37

Let $f : A \to B$ be an injective (one-to-one) function. Define $g : 2^A \to 2^B$ as:
$g(C) = \{f(x) \mid x \in C\}$, for all subsets C of A.
Define $h : 2^B \to 2^A$ as: $h(D) = \{x \mid x \in A, f(x) \in D\}$, for all subsets D of B. Which of the following statements is always true?

A. $g(h(D)) \subseteq D$
B. $g(h(D)) \supseteq D$
C. $g(h(D)) \cap D = \phi$
D. $g(h(D)) \cap (B-D) \neq \phi$

gatecse-2003 set-theory&algebra functions difficult

Answer key

4.3.12 Functions: GATE CSE 2003 | Question: 39

Let $\Sigma = \{a,b,c,d,e\}$ be an alphabet. We define an encoding scheme as follows:
$g(a) = 3, g(b) = 5, g(c) = 7, g(d) = 9, g(e) = 11.$
Let p_i denote the i-th prime number $(p_1 = 2)$.

For a non-empty string $s = a_1 \dots a_n$, where each $a_i \in \Sigma$, define $f(s) = \Pi_{i=1}^{n} P_i^{g(a_i)}$.

For a non-empty sequence$\langle s_j, \dots, s_n \rangle$ of stings from Σ^+, define $h(\langle s_i \dots s_n \rangle) = \Pi_{i=1}^{n} P_i^{f(s_i)}$
Which of the following numbers is the encoding, h, of a non-empty sequence of strings?

A. $2^7 3^7 5^7$ B. $2^8 3^8 5^8$ C. $2^9 3^9 5^9$ D. $2^{10} 3^{10} 5^{10}$

gatecse-2003 set-theory&algebra functions difficult

Answer key

4.3.13 Functions: GATE CSE 2005 | Question: 43

Let $f : B \to C$ and $g : A \to B$ be two functions and let $h = fog$. Given that h is an onto function which one of the following is TRUE?

A. f and g should both be onto functions
B. f should be onto but g need not to be onto
C. g should be onto but f need not be onto
D. both f and g need not be onto

gatecse-2005 set-theory&algebra functions normal

Answer key

4.3.14 Functions: GATE CSE 2006 | Question: 2

Let X, Y, Z be sets of sizes x, y and z respectively. Let $W = X \times Y$ and E be the set of all subsets of W. The number of functions from Z to E is

A. $z^{2^{xy}}$
B. $z \times 2^{xy}$
C. $z^{2^{x+y}}$
D. 2^{xyz}

gatecse-2006 set-theory&algebra normal functions

Answer key

4.3.15 Functions: GATE CSE 2006 | Question: 25

Let $S = \{1, 2, 3, \dots, m\}, m > 3$. Let $X_1, \dots, X_n$ be subsets of S each of size 3. Define a function f from S to the set of natural numbers as, $f(i)$ is the number of sets X_j that contain the element i. That is $f(i) = |\{j \mid i \in X_j\}|$ then $\sum_{i=1}^{m} f(i)$ is:

A. $3m$
B. $3n$
C. $2m+1$
D. $2n+1$

gatecse-2006 set-theory&algebra normal functions

Answer key

4.3.16 Functions: GATE CSE 2007 | Question: 3

What is the maximum number of different Boolean functions involving n Boolean variables?

A. n^2
B. 2^n
C. 2^{2^n}
D. 2^{n^2}

gatecse-2007 combinatory functions normal

Answer key

4.3.17 Functions: GATE CSE 2012 | Question: 37

How many onto (or surjective) functions are there from an n-element $(n \geq 2)$ set to a 2-element set?

A. 2^n
B. $2^n - 1$
C. $2^n - 2$
D. $2(2^n - 2)$

gatecse-2012 set-theory&algebra functions normal

Answer key

4.3.18 Functions: GATE CSE 2014 | Set 1 | Question: 50

Let S denote the set of all functions $f : \{0, 1\}^4 \to \{0, 1\}$. Denote by N the number of functions from S to the set $\{0, 1\}$. The value of $\log_2 \log_2 N$ is _______.

gatecse-2014-set1 set-theory&algebra functions combinatory numerical-answers

Answer key

4.3.19 Functions: GATE CSE 2014 | Set 3 | Question: 2

Let X and Y be finite sets and $f : X \to Y$ be a function. Which one of the following statements is TRUE?

A. For any subsets A and B of X, $|f(A \cup B)| = |f(A)| + |f(B)|$
B. For any subsets A and B of X, $f(A \cap B) = f(A) \cap f(B)$
C. For any subsets A and B of X, $|f(A \cap B)| = \min\{|f(A)|, |f(B)|\}$
D. For any subsets S and T of Y, $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$

gatecse-2014-set3 set-theory&algebra functions normal

Answer key

4.3.20 Functions: GATE CSE 2014 | Set 3 | Question: 49

Consider the set of all functions $f : \{0, 1, \dots, 2014\} \to \{0, 1, \dots, 2014\}$ such that $f(f(i)) = i$, for all $0 \leq i \leq 2014$. Consider the following statements:

P. For each such function it must be the case that for every i, $f(i) = i$.

Q. For each such function it must be the case that for some i, $f(i) = i$.

R. Each function must be onto.

Which one of the following is CORRECT?

A. P, Q and R are true
B. Only Q and R are true
C. Only P and Q are true
D. Only R is true

gatecse-2014-set3 set-theory&algebra functions normal

Answer key

4.3.21 Functions: GATE CSE 2015 | Set 1 | Question: 5

If $g(x) = 1 - x$ and $h(x) = \frac{x}{x-1}$, then $\frac{g(h(x))}{h(g(x))}$ is:

A. $\frac{h(x)}{g(x)}$
B. $\frac{-1}{x}$
C. $\frac{g(x)}{h(x)}$
D. $\frac{x}{(1-x)^2}$

gatecse-2015-set1 set-theory&algebra functions normal

Answer key

4.3.22 Functions: GATE CSE 2015 | Set 2 | Question: 40

The number of onto functions (surjective functions) from set $X = \{1, 2, 3, 4\}$ to set $Y = \{a, b, c\}$ is ______.

gatecse-2015-set2 set-theory&algebra functions normal numerical-answers

Answer key

4.3.23 Functions: GATE CSE 2015 | Set 2 | Question: 54

Let X and Y denote the sets containing 2 and 20 distinct objects respectively and F denote the set of all possible functions defined from X to Y. Let f be randomly chosen from F. The probability of f being one-to-one is ______.

gatecse-2015-set2 set-theory&algebra functions normal numerical-answers

Answer key

4.3.24 Functions: GATE CSE 2016 | Set 1 | Question: 28

A function $f : \mathbb{N}^+ \to \mathbb{N}^+$, defined on the set of positive integers $\mathbb{N}^+$, satisfies the following properties:

$$f(n) = f(n/2) \quad \text{if } n \text{ is even}$$

$$f(n) = f(n+5) \quad \text{if } n \text{ is odd}$$

Let $R = \{i \mid \exists j : f(j) = i\}$ be the set of distinct values that f takes. The maximum possible size of R is _____________.

gatecse-2016-set1 set-theory&algebra functions normal numerical-answers

Answer key

4.3.25 Functions: GATE CSE 2021 | Set 2 | Question: 11

Consider the following sets, where $n \geq 2$:

- S_1: Set of all $n \times n$ matrices with entries from the set $\{a, b, c\}$
- S_2: Set of all functions from the set $\{0, 1, 2, \ldots, n^2 - 1\}$ to the set $\{0, 1, 2\}$

Which of the following choice(s) is/are correct?

A. There does not exist a bijection from S_1 to S_2
B. There exists a surjection from S_1 to S_2
C. There exists a bijection from S_1 to S_2
D. There does not exist an injection from S_1 to S_2

gatecse-2021-set2 multiple-selects set-theory&algebra functions one-mark

Answer key

4.3.26 Functions: GATE CSE 2023 | Question: 39

Let $f : A \to B$ be an onto (or surjective) function, where A and B are nonempty sets. Define an equivalence relation $\sim$ on the set A as

$$a_1 \sim a_2 \text{ if } f(a_1) = f(a_2),$$

where $a_1, a_2 \in A$. Let $\mathcal{E} = \{[x] : x \in A\}$ be the set of all the equivalence classes under $\sim$. Define a new mapping $F : \mathcal{E} \to B$ as

$$F([x]) = f(x), \quad \text{for all the equivalence classes } [x] \text{ in } \mathcal{E}.$$

Which of the following statements is/are TRUE?

A. F is NOT well-defined.
B. F is an onto (or surjective) function.
C. F is a one-to-one (or injective) function.
D. F is a bijective function.

gatecse-2023 set-theory&algebra functions multiple-selects two-marks

Answer key

4.3.27 Functions: GATE CSE 2024 | Set 1 | Question: 22

Let A and B be non-empty finite sets such that there exist one-to-one and onto functions (i) from A to B and (ii) from $A \times A$ to $A \cup B$. The number of possible values of $|\text{A}|$ is _____________.

gatecse-2024-set1 numerical-answers set-theory&algebra functions one-mark

Answer key

4.3.28 Functions: GATE DA 2026 | Question: 23

The number of bijections $f(\cdot)$ from the set $S = \{1, 2, 3, 4\}$ to itself such that $f(f(n)) = n$, for all $n \in S$, is _________. (Answer in integer)

gateda-2026 set-theory&algebra functions numerical-answers one-mark

Answer key

4.3.29 Functions: GATE IT 2005 | Question: 31

Let f be a function from a set A to a set B, g a function from B to C, and h a function from A to C, such that $h(a) = g(f(a))$ for all $a \in A$. Which of the following statements is always true for all such functions f and g?

A. g is onto $\implies$ h is onto
B. h is onto $\implies$ f is onto
C. h is onto $\implies$ g is onto
D. h is onto $\implies$ f and g are onto

gateit-2005 set-theory&algebra functions normal

Answer key

4.3.30 Functions: GATE IT 2006 | Question: 6

Given a boolean function $f(x_1, x_2, \dots, x_n)$, which of the following equations is NOT true?

A. $f(x_1, x_2, \dots, x_n) = x_1' f(x_1, x_2, \dots, x_n) + x_1 f(x_1, x_2, \dots, x_n)$
B. $f(x_1, x_2, \dots, x_n) = x_2 f(x_1, x_2, \dots, x_n) + x_2' f(x_1, x_2, \dots, x_n)$
C. $f(x_1, x_2, \dots, x_n) = x_n' f(x_1, x_2, \dots, 0) + x_n f(x_1, x_2, \dots, 1)$
D. $f(x_1, x_2, \dots, x_n) = f(0, x_2, \dots, x_n) + f(1, x_2, \dots, x_n)$

gateit-2006 set-theory&algebra functions normal

Answer key

4.4 Group Theory (33)

4.4.1 Group Theory: GATE CSE 1988 | Question: 2xviii

Show that if G is a group such that $(a * b)^2 = a^2 * b^2$ for all a, b belonging to G, then G is an abelian.

gate1988 descriptive group-theory

Answer key

4.4.2 Group Theory: GATE CSE 1990 | Question: 2-x

Match the pairs in the following questions:

(a)	Groups	(p)	Associativity
(b)	Semigroups	(q)	Identity
(c)	Monoids	(r)	Commutativity
(d)	Abelian groups	(s)	Left inverse

gate1990 match-the-following set-theory&algebra group-theory easy

Answer key

4.4.3 Group Theory: GATE CSE 1992 | Question: 14a

If G is a group of even order, then show that there exists an element $a \neq e$, the identity in G, such that $a^2 = e$.

gate1992 set-theory&algebra group-theory normal descriptive proof

Answer key

4.4.4 Group Theory: GATE CSE 1993 | Question: 28

Let $(\{p, q\}, *)$ be a semigroup where $p * p = q$. Show that:

a. $p * q = q * p$ and
b. $q * q = q$